Red Hat OpenShift Data Foundation 4.21

Monitoring OpenShift Data Foundation

View cluster health, metrics, or set alerts.

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Abstract

Read this document for instructions on monitoring Red Hat OpenShift Data Foundation using the Block and File, and Object dashboards.

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CHAPTER 1. CLUSTER HEALTH

1.1. VERIFYING OPENSHIFT DATA FOUNDATION IS HEALTHY

Storage health is visible on the Block and File and Object dashboards.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. Check if the **Status** card has a green tick in the **Block and File** and the **Object** tabs.
   Green tick indicates that the cluster is healthy.

See Section 1.2, “Storage health levels and cluster state” for information about the different health states and the alerts that appear.

1.2. STORAGE HEALTH LEVELS AND CLUSTER STATE

Status information and alerts related to OpenShift Data Foundation are displayed in the storage dashboards.

1.2.1. Block and File dashboard indicators

The **Block and File** dashboard shows the complete state of OpenShift Data Foundation and the state of persistent volumes.

The states that are possible for each resource type are listed in the following table.

Table 1.1. OpenShift Data Foundation health levels

State	Icon	Description
UNKNOWN		OpenShift Data Foundation is not deployed or unavailable.
Green Tick		Cluster health is good.
Warning		OpenShift Data Foundation cluster is in a warning state. In internal mode, an alert will be displayed along with the issue details. Alerts are not displayed for external mode.

State	Icon	Description
Error		OpenShift Data Foundation cluster has encountered an error and some component is nonfunctional. In internal mode, an alert is displayed along with the issue details. Alerts are not displayed for external mode.

1.2.2. Object dashboard indicators

The **Object** dashboard shows the state of the Multicloud Object Gateway and any object claims in the cluster.

The states that are possible for each resource type are listed in the following table.

Table 1.2. Object Service health levels

State	Description
Green Tick	Object storage is healthy.
Multicloud Object Gateway is not running	Shown when NooBaa system is not found.
All resources are unhealthy	Shown when all NooBaa pools are unhealthy.
Many buckets have issues	Shown when >= 50% of buckets encounter error(s).
Some buckets have issues	Shown when >= 30% of buckets encounter error(s).
Unavailable	Shown when network issues and/or errors exist.

1.2.3. Alert panel

The Alert panel appears below the **Status** card in both the **Block and File** dashboard and the **Object** dashboard when the cluster state is not healthy.

Information about specific alerts and how to respond to them is available in Troubleshooting OpenShift Data Foundation.

1.3. VIEWING OPENSHIFT DATA FOUNDATION INFRASTRUCTURE HEALTH

This section details how to monitor the overall health of your infrastructure and temporarily disable health checks when needed.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.

2. In **Overview** tab, view the **Infrastructure health card**. Here you will see your infrastructure health score out of 100%. This score reflects the overall operational status of your OpenShift Data Foundation cluster. Your system starts with a health score of 100%. Points are deducted whenever issues are found. The amount deducted depends on the severity of the issue:

   - **Minor**: subtract 2 points
   - **Medium**: subtract 10 points
   - **Critical**: subtract 20 points

View health checks and alerts

1. In the **Infrastructure health card**, click **View health checks**. Here you will see any of the following alerts:

Table 1.3. Infrastructure health checks

Health check	Severity	Description	Metrics	Notes / Requirements
Node Latency	Medium	Maximum latency between nodes running OSDs should be < 10ms roundtrip (last 24h). Maximum latency between other ODF nodes should be < 100ms roundtrip (last 24h).	**probe_icmp_duration_seconds**	Requires Prometheus Blackbox Exporter to be deployed and configured.
MTU Check	Minor	Verify that network interfaces used by ODF nodes have an MTU < 9000 bytes.	**node_network_mtu_bytes**	
Disk Busy	Medium	Checks if disks used by ODF are > 90% busy on average.	**node_disk_io_time_seconds_total**	
ODF Pod Restarts	Medium	Detect if any core ODF pods have restarted within the last 24 hours.	**kube_pod_container_status_restarts_total**	

Health check	Severity	Description	Metrics	Notes / Requirements
NIC Bandwidth Saturation	Medium	Detect if any ODF node NIC used for cluster traffic exceeded 90% bandwidth utilization over a 5 minute period within the last 24 hours.	• **node_network_receive_bytes_total** • **node_network_transmit_bytes_total** • **node_network_speed_bytes**	

Table 1.4. Infrastructure health alerts

Alert	Severity
ODFNodeLatencyHighOnOSDNodes	Medium
ODFNodeLatencyHighOnNonOSDNodes	Medium
ODFNodeMTULessThan9000	Minor
ODFDiskUtilizationHigh	Medium
ODFCorePodRestarted	Medium
ODFNodeNICBandwidthSaturation	Medium

[Optional] silence health checks

1. In the **Infrastructure health card**, click **View health checks**.
2. View the list of health checks and select the checkbox next the health check(s) you want to silence.
3. Click **Silence** at the end of the row.
4. Enter a **Duration** and click **Silence**.

CHAPTER 2. MULTICLUSTER STORAGE HEALTH

To view the overall storage health status across all the clusters with OpenShift Data Foundation and manage its capacity, you must first enable the multicluster dashboard on the Hub cluster.

2.1. ENABLING MULTICLUSTER DASHBOARD ON HUB CLUSTER

You can enable the multicluster dashboard on the install screen either before or after installing ODF Multicluster Orchestrator with the console plugin.

Prerequisites

- Ensure that you have installed OpenShift Container Platform version 4.17 and have administrator privileges.
- Ensure that you have installed Multicluster Orchestrator 4.17 operator with plugin for console enabled.
- Ensure that you have installed Red Hat Advanced Cluster Management for Kubernetes (RHACM) 2.11 from Software Catalog. For instructions on how to install, see Installing RHACM.
- Ensure you have enabled observability on RHACM. See Enabling observability guidelines.

Procedure

1. Create the configmap file named **observability-metrics-custom-allowlist.yaml** and add the name of the custom metric to the **metrics_list.yaml** parameter.
   You can use the following YAML to list the OpenShift Data Foundation metrics on Hub cluster. For details, see Adding custom metrics.

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: observability-metrics-custom-allowlist
  Namespace: open-cluster-management-observability
data:
  metrics_list.yaml: |
    names:
      - odf_system_health_status
      - odf_system_map
      - odf_system_raw_capacity_total_bytes
      - odf_system_raw_capacity_used_bytes
    matches:
      - __name__="csv_succeeded",exported_namespace="openshift-storage",name=~"odf-operator.*"
```

2. Run the following command in the **open-cluster-management-observability** namespace:

```
# oc apply -n open-cluster-management-observability -f observability-metrics-custom-allowlist.yaml
```

   After **observability-metrics-custom-allowlist** yaml is created, RHACM will start collecting the listed OpenShift Data Foundation metrics from all the managed clusters.

If you want to exclude specific managed clusters from collecting the observability data, add the following cluster label to your clusters: **observability: disabled**.

3. To view the multicluster health, see chapter verifying multicluster storage dashboard .

2.2. VERIFYING MULTICLUSTER STORAGE HEALTH ON HUB CLUSTER

Prerequisites

Ensure that you have enabled multicluster monitoring. For instructions, see chapter Enabling multicluster dashboard.

Procedure

1. In the OpenShift web console of Hub cluster, ensure **All Clusters** is selected.
2. Navigate to **Data Services** and click **Storage System**.
3. On the Overview tab, verify that there are green ticks in front of **OpenShift Data Foundation** and **Systems**. This indicates that the operator is running and all storage systems are available.
4. In the Status card,
    a. Click **OpenShift Data Foundation** to view the operator status.
    b. Click **Systems** to view the storage system status.
5. The **Storage system capacity** card shows the following details:
    - Name of the storage system
    - Cluster name
    - Graphical representation of total and used capacity in percentage
    - Actual values for total and used capacity in TiB

CHAPTER 3. METRICS

3.1. METRICS IN THE BLOCK AND FILE DASHBOARD

You can navigate to the Block and File dashboard in the OpenShift Web Console as follows:

1. Click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. Click the **Block and File** tab.

The following cards on the Block and File dashboard provides the metrics based on the deployment mode (internal or external):

Details card

The Details card shows the following:

- Service name
- Cluster name
- The name of the Provider on which the system runs (example: **AWS**, **VSphere**, **None** for Bare metal)
- Mode (deployment mode as either Internal or External)
- OpenShift Data Foundation operator version.
- In-transit encryption (shows whether the encryption is enabled or disabled)

Storage Efficiency card

This card shows the compression ratio that represents a compressible data effectiveness metric, which includes all the compression-enabled pools. This card also shows the savings metric that represents the actual disk capacity saved, which includes all the compression-enabled pools and associated replicas.

Inventory card

The Inventory card shows the total number of active Nodes, PersistentVolumeClaims, and PersistentVolumes backed by OpenShift Data Foundation provisioner.

NOTE

For external mode, the number of nodes will be 0 by default as there are no dedicated nodes for OpenShift Data Foundation.

Status card

This card shows whether the cluster is up and running without any errors or is experiencing some issues.
For internal mode, Data Resiliency indicates the status of data re-balancing in Ceph across the replicas. When the internal mode cluster is in a warning or error state, the Alerts section is shown along with the relevant alerts.

For external mode, Data Resiliency and alerts are not displayed

Raw Capacity card

This card shows the total raw storage capacity which includes replication on the cluster.

- **Used** legend indicates space used raw storage capacity on the cluster
- **Available** legend indicates the available raw storage capacity on the cluster.

NOTE

This card is not applicable for external mode clusters.

Consumption trend card

This card shows the storage consumption rate in GiB per day which is calculated based on the actual capacity utilization, historical usage, and current consumption rate. Also, the card shows the number of days left for the storage to reach the threshold capacity.

Requested Capacity

This card shows the actual amount of non-replicated data stored in the cluster and its distribution. You can choose between Projects, Storage Classes, Pods, and Persistent Volume Claims from the drop-down list on the top of the card. You need to select a namespace for the Persistent Volume Claims option. These options are for filtering the data shown in the graph. The graph displays the requested capacity for only the top five entities based on usage. The aggregate requested capacity of the remaining entities is displayed as **Other**.

Option	Display
Projects	The aggregated capacity of each project which is using the OpenShift Data Foundation and how much is being used.
Storage Classes	The aggregate capacity which is based on the OpenShift Data Foundation based storage classes.
Pods	All the pods that are trying to use the PVC that are backed by OpenShift Data Foundation provisioner.
PVCs	All the PVCs in the namespace that you selected from the dropdown list and that are mounted on to an active pod. PVCs that are not attached to pods are not included.

Utilization card

The card shows used capacity, input/output operations per second, latency, throughput, and recovery information for the internal mode cluster.
For external mode, this card shows only the used and requested capacity details for that cluster.

Activity card

This card shows the current and the past activities of the OpenShift Data Foundation cluster. The card is separated into two sections:

- **Ongoing**: Displays the progress of ongoing activities related to rebuilding of data resiliency and upgrading of OpenShift Data Foundation operator.
- **Recent Events**: Displays the list of events that happened in the **openshift-storage** namespace.

3.2. METRICS IN THE OBJECT DASHBOARD

You can navigate to the Object dashboard in the OpenShift Web Console as follows:

1. Click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. Click the **Object** tab.

The following metrics are available in the Object dashboard:

Details card

This card shows the following information:

- **Service Name**: The Multicloud Object Gateway (MCG) service name.
- **System Name**: The Multicloud Object Gateway and RADOS Object Gateway system names. The Multicloud Object Gateway system name is also a hyperlink to the MCG management user interface.
- **Provider**: The name of the provider on which the system runs (example: **AWS**, **VSphere**, **None** for Bare metal)
- **Version**: OpenShift Data Foundation operator version.

Storage Efficiency card

In this card you can view how the MCG optimizes the consumption of the storage backend resources through deduplication and compression and provides you with a calculated efficiency ratio (application data vs logical data) and an estimated savings figure (how many bytes the MCG did not send to the storage provider) based on capacity of bare metal and cloud based storage and egress of cloud based storage.

Buckets card

Buckets are containers maintained by the MCG and RADOS Object Gateway to store data on behalf of the applications. These buckets are created and accessed through object bucket claims (OBCs). A specific policy can be applied to bucket to customize data placement, data spill-over, data resiliency, capacity quotas, and so on.
In this card, information about object buckets (OB) and object bucket claims (OBCs) is shown separately. OB includes all the buckets that are created using S3 or the user interface(UI) and OBC includes all the buckets created using YAMLs or the command line interface (CLI). The number displayed on the left of the bucket type is the total count of OBs or OBCs. The number displayed on the right shows the error count and is visible only when the error count is greater than zero. You can click on the number to see the list of buckets that has the warning or error status.

Resource Providers card

This card displays a list of all Multicloud Object Gateway and RADOS Object Gateway resources that are currently in use. Those resources are used to store data according to the buckets policies and can be a cloud-based resource or a bare metal resource.

Status card

This card shows whether the system and its services are running without any issues. When the system is in a warning or error state, the alerts section is shown and the relevant alerts are displayed there. Click the alert links beside each alert for more information about the issue. For information about health checks, see Cluster health.
If multiple object storage services are available in the cluster, click the service type (such as **Object Service** or **Data Resiliency**) to see the state of the individual services.

Data resiliency in the status card indicates if there is any resiliency issue regarding the data stored through the Multicloud Object Gateway and RADOS Object Gateway.

Capacity breakdown card

In this card you can visualize how applications consume the object storage through the Multicloud Object Gateway and RADOS Object Gateway. You can use the **Service Type** drop-down to view the capacity breakdown for the Multicloud Gateway and Object Gateway separately. When viewing the Multicloud Object Gateway, you can use the **Break By** drop-down to filter the results in the graph by either **Projects** or **Bucket Class**.

Performance card

In this card, you can view the performance of the Multicloud Object Gateway or RADOS Object Gateway. Use the **Service Type** drop-down to choose which you would like to view.
For Multicloud Object Gateway accounts, you can view the I/O operations and logical used capacity. For providers, you can view I/O operation, physical and logical usage, and egress.

The following tables explain the different metrics that you can view based on your selection from the drop-down menus on the top of the card:

Table 3.1. Indicators for Multicloud Object Gateway

Consumer types	Metrics	Chart display
Accounts	I/O operations	Displays read and write I/O operations for the top five consumers. The total reads and writes of all the consumers is displayed at the bottom. This information helps you monitor the throughput demand (IOPS) per application or account.
Accounts	Logical Used Capacity	Displays total logical usage of each account for the top five consumers. This helps you monitor the throughput demand per application or account.

Consumer types	Metrics	Chart display
Providers	I/O operations	Displays the count of I/O operations generated by the MCG when accessing the storage backend hosted by the provider. This helps you understand the traffic in the cloud so that you can improve resource allocation according to the I/O pattern, thereby optimizing the cost.
Providers	Physical vs Logical usage	Displays the data consumption in the system by comparing the physical usage with the logical usage per provider. This helps you control the storage resources and devise a placement strategy in line with your usage characteristics and your performance requirements while potentially optimizing your costs.
Providers	Egress	The amount of data the MCG retrieves from each provider (read bandwidth originated with the applications). This helps you understand the traffic in the cloud to improve resource allocation according to the egress pattern, thereby optimizing the cost.

For the RADOS Object Gateway, you can use the **Metric** drop-down to view the **Latency** or **Bandwidth**.

- **Latency**: Provides a visual indication of the average GET/PUT latency imbalance across RADOS Object Gateway instances.
- **Bandwidth**: Provides a visual indication of the sum of GET/PUT bandwidth across RADOS Object Gateway instances.

Activity card

This card displays what activities are happening or have recently happened in the OpenShift Data Foundation cluster. The card is separated into two sections:

- **Ongoing**: Displays the progress of ongoing activities related to rebuilding of data resiliency and upgrading of OpenShift Data Foundation operator.
- **Recent Events**: Displays the list of events that happened in the **openshift-storage** namespace.

3.3. POOL METRICS

The Pool metrics dashboard provides information to ensure efficient data consumption, and how to enable or disable compression if less effective.

Viewing pool metrics

To view the pool list:

1. Click **Storage → Data Foundation**
2. In the **Storage systems** tab, select the storage system and then click **BlockPools**.

When you click on a pool name, the following cards on each Pool dashboard is displayed along with the metrics based on the deployment mode (internal or external):

Details card

The Details card shows the following:

- Pool Name
- Volume type
- Replicas

Status card

This card shows whether the pool is up and running without any errors or is experiencing some issues.

Mirroring card

When the mirroring option is enabled, this card shows the mirroring status, image health, and last checked time-stamp. The mirroring metrics are displayed when cluster level mirroring is enabled. The metrics help to prevent disaster recovery failures and notify of any discrepancies so that the data is kept intact.
The mirroring card shows high-level information such as:

- Mirroring state as either enabled or disabled for the particular pool.
- Status of all images under the pool as replicating successfully or not.
- Percentage of images that are replicating and not replicating.

Inventory card

The Inventory card shows the number of storage classes and Persistent Volume Claims.

Compression card

This card shows the compression status as enabled or disabled as the case may be. It also displays the storage efficiency details as follows:

- Compression eligibility that indicates what portion of written compression-eligible data is compressible (per ceph parameters)
- Compression ratio of compression-eligible data
- Compression savings provides the total savings (including replicas) of compression-eligible data

For information on how to enable or disable compression for an existing pool, see Updating an existing pool.

Raw Capacity card

This card shows the total raw storage capacity which includes replication, on the cluster.

- **Used** legend indicates storage capacity used by the pool
- **Available** legend indicates the available raw storage capacity on the cluster

Performance card

In this card, you can view the usage of I/O operations and throughput demand per application or account. The graph indicates the average latency or bandwidth across the instances.

3.4. NETWORK FILE SYSTEM METRICS

The Network File System (NFS) metrics dashboard provides enhanced observability for NFS mounts such as the following:

- Mount point for any exported NFS shares
- Number of client mounts
- A breakdown statistics of the clients that are connected to help determine internal versus the external client mounts
- Grace period status of the Ganesha server
- Health statuses of the Ganesha server

Prerequisites

- OpenShift Container Platform is installed and you have administrative access to OpenShift Web Console.
- Ensure that NFS is enabled.

Procedure

You can navigate to the **Network file system** dashboard in the OpenShift Web Console as follows:

1. Click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. Click the **Network file system** tab.
   This tab is available only when NFS is enabled.

NOTE

When you enable or disable NFS from command-line interface, you must perform hard refresh to display or hide the **Network file system** tab in the dashboard.

The following NFS metrics are displayed:

Status Card
: This card shows the status of the server based on the total number of active worker threads. Non-zero threads specify healthy status.

Throughput Card
: This card shows the throughput of the server which is the summation of the total request bytes and total response bytes for both read and write operations of the server.

Top client Card
: This card shows the throughput of clients which is the summation of the total of the response bytes sent by a client and the total request bytes by a client for both read and write operations. It shows the top three of such clients.

3.5. ENABLING METADATA ON RBD AND CEPHFS VOLUMES

You can set the persistent volume claim (PVC), persistent volume (PV), and Namespace names in the RADOS block device (RBD) and CephFS volumes for monitoring purposes. This enables you to read the RBD and CephFS metadata to identify the mapping between the OpenShift Container Platform and RBD and CephFS volumes.

To enable RADOS block device (RBD) and CephFS volume metadata feature, you need to set the **CSI_ENABLE_METADATA** variable in the **rook-ceph-operator-config configmap**. By default, this feature is disabled. If you enable the feature after upgrading from a previous version, the existing PVCs will not contain the metadata. Also, when you enable the metadata feature, the PVCs that were created before enabling will not have the metadata.

Prerequisites

- Ensure to install **ocs_operator** and create a **storagecluster** for the operator.
- Ensure that the **storagecluster** is in **Ready** state.

```
$ oc get storagecluster
NAME                 AGE   PHASE   EXTERNAL   CREATED AT             VERSION
ocs-storagecluster   57m   Ready              2022-08-30T06:52:58Z   4.12.0
```

Procedure

1. Edit the **rook-ceph** operator **ConfigMap** to mark **CSI_ENABLE_METADATA** to **true**.

```
$ oc patch cm rook-ceph-operator-config -n openshift-storage -p $'data:\n
"CSI_ENABLE_METADATA":  "true"'
configmap/rook-ceph-operator-config patched
```

2. Wait for the respective CSI CephFS plugin provisioner pods and CSI RBD plugin pods to reach the **Running** state.

> **NOTE**
>
> Ensure that the **setmetadata** variable is automatically set after the metadata feature is enabled. This variable should not be available when the metadata feature is disabled.

```
$ oc get pods | grep csi

openshift-storage.cephfs.csi.ceph.com-ctrlplugin-65986bb74nznxn   7/7     Running     0
56m
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-65986bb74zbf45   7/7     Running     0
56m
openshift-storage.cephfs.csi.ceph.com-nodeplugin-bzvlt          3/3     Running     0      56m
openshift-storage.cephfs.csi.ceph.com-nodeplugin-w5l8s           3/3     Running     0      56m
openshift-storage.cephfs.csi.ceph.com-nodeplugin-xgtj2          3/3     Running     0      56m
openshift-storage.rbd.csi.ceph.com-ctrlplugin-c8cbc88b-rx4ff     3/3     Running     0      56m
openshift-storage.rbd.csi.ceph.com-ctrlplugin-c8cbc88b-zq6rl     3/3     Running     0      56m
openshift-storage.rbd.csi.ceph.com-nodeplugin-8pvn8             3/3     Running     0      56m
openshift-storage.rbd.csi.ceph.com-nodeplugin-mgjvh             6/6     Running     0      56m
openshift-storage.rbd.csi.ceph.com-nodeplugin-xczkv             6/6     Running     0      56m
```

Verification steps

- To verify the metadata for RBD PVC:

  1. Create a PVC.

     ```
     $ cat <<EOF | oc create -f -
     apiVersion: v1
     kind: PersistentVolumeClaim
     metadata:
       name: rbd-pvc
     spec:
       accessModes:
         - ReadWriteOnce
       resources:
         requests:
           storage: 1Gi
       storageClassName: ocs-storagecluster-ceph-rbd
     EOF
     ```

  2. Check the status of the PVC.

     ```
     $ oc get pvc | grep rbd-pvc
     rbd-pvc                           Bound    pvc-30628fa8-2966-499c-832d-a6a3a8ebc594   1Gi
     RWO            ocs-storagecluster-ceph-rbd   32s
     ```

  3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
     For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

     ```
     [sh-4.x]$ rbd ls ocs-storagecluster-cephblockpool

     csi-vol-7d67bfad-2842-11ed-94bd-0a580a830012
     csi-vol-ed5ce27b-2842-11ed-94bd-0a580a830012

     [sh-4.x]$ rbd image-meta ls ocs-storagecluster-cephblockpool/csi-vol-ed5ce27b-2842-
     11ed-94bd-0a580a830012
     ```

     There are four metadata on this image:

```
Key                              Value
csi.ceph.com/cluster/name        6cd7a18d-7363-4830-ad5c-f7b96927f026
csi.storage.k8s.io/pv/name       pvc-30628fa8-2966-499c-832d-a6a3a8ebc594
csi.storage.k8s.io/pvc/name      rbd-pvc
csi.storage.k8s.io/pvc/namespace  openshift-storage
```

- To verify the metadata for RBD clones:

1. Create a clone.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: rbd-pvc-clone
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  dataSource:
    name: rbd-pvc
    kind: PersistentVolumeClaim
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
EOF
```

2. Check the status of the clone.

```
$ oc get pvc | grep rbd-pvc
rbd-pvc                         Bound    pvc-30628fa8-2966-499c-832d-a6a3a8ebc594   1Gi
RWO            ocs-storagecluster-ceph-rbd   15m
rbd-pvc-clone                       Bound    pvc-0d72afda-f433-4d46-a7f1-a5fcb3d766e0   1Gi
RWO            ocs-storagecluster-ceph-rbd   52s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
[sh-4.x]$ rbd ls ocs-storagecluster-cephblockpool
csi-vol-063b982d-2845-11ed-94bd-0a580a830012
csi-vol-063b982d-2845-11ed-94bd-0a580a830012-temp
csi-vol-7d67bfad-2842-11ed-94bd-0a580a830012
csi-vol-ed5ce27b-2842-11ed-94bd-0a580a830012

[sh-4.x]$ rbd image-meta ls ocs-storagecluster-cephblockpool/csi-vol-063b982d-2845-
11ed-94bd-0a580a830012
There are 4 metadata on this image:

Key                              Value
csi.ceph.com/cluster/name        6cd7a18d-7363-4830-ad5c-f7b96927f026
csi.storage.k8s.io/pv/name       pvc-0d72afda-f433-4d46-a7f1-a5fcb3d766e0
csi.storage.k8s.io/pvc/name      rbd-pvc-clone
csi.storage.k8s.io/pvc/namespace  openshift-storage
```

- To verify the metadata for RBD Snapshots:

1. Create a snapshot.

```
$ cat <<EOF | oc create -f -
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
  name: rbd-pvc-snapshot
spec:
  volumeSnapshotClassName: ocs-storagecluster-rbdplugin-snapclass
  source:
    persistentVolumeClaimName: rbd-pvc
EOF
volumesnapshot.snapshot.storage.k8s.io/rbd-pvc-snapshot created
```

2. Check the status of the snapshot.

```
$ oc get volumesnapshot
NAME             READYTOUSE   SOURCEPVC   SOURCESNAPSHOTCONTENT
RESTORESIZE   SNAPSHOTCLASS                            SNAPSHOTCONTENT
CREATIONTIME   AGE
rbd-pvc-snapshot   true         rbd-pvc                                  1Gi          ocs-storagecluster-
rbdplugin-snapclass   snapcontent-b992b782-7174-4101-8fe3-e6e478eb2c8f   17s
18s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
[sh-4.x]$ rbd ls ocs-storagecluster-cephblockpool
csi-snap-a1e24408-2848-11ed-94bd-0a580a830012
csi-vol-063b982d-2845-11ed-94bd-0a580a830012
csi-vol-063b982d-2845-11ed-94bd-0a580a830012-temp
csi-vol-7d67bfad-2842-11ed-94bd-0a580a830012
csi-vol-ed5ce27b-2842-11ed-94bd-0a580a830012

[sh-4.x]$ rbd image-meta ls ocs-storagecluster-cephblockpool/csi-snap-a1e24408-2848-
11ed-94bd-0a580a830012
There are 4 metadata on this image:

Key                                           Value
csi.ceph.com/cluster/name                     6cd7a18d-7363-4830-ad5c-f7b96927f026
csi.storage.k8s.io/volumesnapshot/name        rbd-pvc-snapshot
csi.storage.k8s.io/volumesnapshot/namespace   openshift-storage
csi.storage.k8s.io/volumesnapshotcontent/name  snapcontent-b992b782-7174-4101-
8fe3-e6e478eb2c8f
```

- Verify the metadata for RBD Restore:

1. Restore a volume snapshot.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: PersistentVolumeClaim
```

```
metadata:
  name: rbd-pvc-restore
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  dataSource:
    name: rbd-pvc-snapshot
    kind: VolumeSnapshot
    apiGroup: snapshot.storage.k8s.io
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
EOF
persistentvolumeclaim/rbd-pvc-restore created
```

2. Check the status of the restored volume snapshot.

```
$ oc get pvc | grep rbd
db-noobaa-db-pg-0            Bound    pvc-615e2027-78cd-4ea2-a341-fdedd50c5208
50Gi       RWO          ocs-storagecluster-ceph-rbd   51m
rbd-pvc                      Bound    pvc-30628fa8-2966-499c-832d-a6a3a8ebc594   1Gi
RWO          ocs-storagecluster-ceph-rbd   47m
rbd-pvc-clone                Bound    pvc-0d72afda-f433-4d46-a7f1-a5fcb3d766e0   1Gi
RWO          ocs-storagecluster-ceph-rbd   32m
rbd-pvc-restore              Bound    pvc-f900e19b-3924-485c-bb47-01b84c559034   1Gi
RWO          ocs-storagecluster-ceph-rbd   111s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
[sh-4.x]$ rbd ls ocs-storagecluster-cephblockpool
csi-snap-a1e24408-2848-11ed-94bd-0a580a830012
csi-vol-063b982d-2845-11ed-94bd-0a580a830012
csi-vol-063b982d-2845-11ed-94bd-0a580a830012-temp
csi-vol-5f6e0737-2849-11ed-94bd-0a580a830012
csi-vol-7d67bfad-2842-11ed-94bd-0a580a830012
csi-vol-ed5ce27b-2842-11ed-94bd-0a580a830012

[sh-4.x]$ rbd image-meta ls ocs-storagecluster-cephblockpool/csi-vol-5f6e0737-2849-
11ed-94bd-0a580a830012
There are 4 metadata on this image:

Key                           Value
csi.ceph.com/cluster/name        6cd7a18d-7363-4830-ad5c-f7b96927f026
csi.storage.k8s.io/pv/name       pvc-f900e19b-3924-485c-bb47-01b84c559034
csi.storage.k8s.io/pvc/name      rbd-pvc-restore
csi.storage.k8s.io/pvc/namespace  openshift-storage
```

- To verify the metadata for CephFS PVC:

1. Create a PVC.

```
cat <<EOF | oc create -f -
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: cephfs-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: ocs-storagecluster-cephfs
EOF
```

2. Check the status of the PVC.

```
oc get pvc | grep cephfs
cephfs-pvc                    Bound    pvc-4151128c-86f0-468b-b6e7-5fdfb51ba1b9   1Gi
RWO            ocs-storagecluster-cephfs     11s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
   For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
$ ceph fs volume ls
[
    {
        "name": "ocs-storagecluster-cephfilesystem"
    }
]

$ ceph fs subvolumegroup ls ocs-storagecluster-cephfilesystem
[
    {
        "name": "csi"
    }
]

$ ceph fs subvolume ls ocs-storagecluster-cephfilesystem --group_name csi
[
    {
        "name": "csi-vol-25266061-284c-11ed-95e0-0a580a810215"
    }
]

$ ceph fs subvolume metadata ls ocs-storagecluster-cephfilesystem csi-vol-25266061-
284c-11ed-95e0-0a580a810215 --group_name=csi --format=json

{
    "csi.ceph.com/cluster/name": "6cd7a18d-7363-4830-ad5c-f7b96927f026",
    "csi.storage.k8s.io/pv/name": "pvc-4151128c-86f0-468b-b6e7-5fdfb51ba1b9",
    "csi.storage.k8s.io/pvc/name": "cephfs-pvc",
    "csi.storage.k8s.io/pvc/namespace": "openshift-storage"
}
```

- To verify the metadata for CephFS clone:

1. Create a clone.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: cephfs-pvc-clone
spec:
  storageClassName: ocs-storagecluster-cephfs
  dataSource:
    name: cephfs-pvc
    kind: PersistentVolumeClaim
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 1Gi
EOF
persistentvolumeclaim/cephfs-pvc-clone created
```

2. Check the status of the clone.

```
$ oc get pvc | grep cephfs
cephfs-pvc                    Bound    pvc-4151128c-86f0-468b-b6e7-5fdfb51ba1b9   1Gi
RWO           ocs-storagecluster-cephfs     9m5s
cephfs-pvc-clone              Bound    pvc-3d4c4e78-f7d5-456a-aa6e-4da4a05ca4ce   1Gi
RWX           ocs-storagecluster-cephfs     20s
```

3. Verify the metadata using the in the Red Hat Ceph Storage command-line interface (CLI). For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
[rook@rook-ceph-tools-c99fd8dfc-6sdbg /]$ ceph fs subvolume ls ocs-storagecluster-
cephfilesystem --group_name csi
[
    {
        "name": "csi-vol-5ea23eb0-284d-11ed-95e0-0a580a810215"
    },
    {
        "name": "csi-vol-25266061-284c-11ed-95e0-0a580a810215"
    }
]

[rook@rook-ceph-tools-c99fd8dfc-6sdbg /]$ ceph fs subvolume metadata ls ocs-
storagecluster-cephfilesystem csi-vol-5ea23eb0-284d-11ed-95e0-0a580a810215 --
group_name=csi --format=json

{
    "csi.ceph.com/cluster/name": "6cd7a18d-7363-4830-ad5c-f7b96927f026",
    "csi.storage.k8s.io/pv/name": "pvc-3d4c4e78-f7d5-456a-aa6e-4da4a05ca4ce",
    "csi.storage.k8s.io/pvc/name": "cephfs-pvc-clone",
    "csi.storage.k8s.io/pvc/namespace": "openshift-storage"
}
```

- To verify the metadata for CephFS volume snapshot:

1. Create a volume snapshot.

```
$ cat <<EOF | oc create -f -
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
  name: cephfs-pvc-snapshot
spec:
  volumeSnapshotClassName: ocs-storagecluster-cephfsplugin-snapclass
  source:
    persistentVolumeClaimName: cephfs-pvc
EOF
volumesnapshot.snapshot.storage.k8s.io/cephfs-pvc-snapshot created
```

2. Check the status of the volume snapshot.

```
$ oc get volumesnapshot
NAME                READYTOUSE   SOURCEPVC    SOURCESNAPSHOTCONTENT
RESTORESIZE   SNAPSHOTCLASS                              SNAPSHOTCONTENT
CREATIONTIME   AGE
cephfs-pvc-snapshot   true         cephfs-pvc                              1Gi           ocs-
storagecluster-cephfsplugin-snapclass   snapcontent-f0f17463-d13b-4e13-b44e-
6340bbb3bee0   9s             9s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
   For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
$ ceph fs subvolume snapshot ls ocs-storagecluster-cephfilesystem csi-vol-25266061-
284c-11ed-95e0-0a580a810215 --group_name csi
[
    {
        "name": "csi-snap-06336f4e-284e-11ed-95e0-0a580a810215"
    }
]

$ ceph fs subvolume snapshot metadata ls ocs-storagecluster-cephfilesystem csi-vol-
25266061-284c-11ed-95e0-0a580a810215 csi-snap-06336f4e-284e-11ed-95e0-
0a580a810215 --group_name=csi --format=json

{
    "csi.ceph.com/cluster/name": "6cd7a18d-7363-4830-ad5c-f7b96927f026",
    "csi.storage.k8s.io/volumesnapshot/name": "cephfs-pvc-snapshot",
    "csi.storage.k8s.io/volumesnapshot/namespace": "openshift-storage",
    "csi.storage.k8s.io/volumesnapshotcontent/name": "snapcontent-f0f17463-d13b-4e13-
b44e-6340bbb3bee0"
}
```

- To verify the metadata of the CephFS Restore:

1. Restore a volume snapshot.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: PersistentVolumeClaim
```

```
metadata:
  name: cephfs-pvc-restore
spec:
  storageClassName: ocs-storagecluster-cephfs
  dataSource:
    name: cephfs-pvc-snapshot
    kind: VolumeSnapshot
    apiGroup: snapshot.storage.k8s.io
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 1Gi
EOF
persistentvolumeclaim/cephfs-pvc-restore created
```

2. Check the status of the restored volume snapshot.

```
$ oc get pvc | grep cephfs
cephfs-pvc                    Bound    pvc-4151128c-86f0-468b-b6e7-5fdfb51ba1b9   1Gi
RWO            ocs-storagecluster-cephfs     29m
cephfs-pvc-clone              Bound    pvc-3d4c4e78-f7d5-456a-aa6e-4da4a05ca4ce   1Gi
RWX            ocs-storagecluster-cephfs     20m
cephfs-pvc-restore            Bound    pvc-43d55ea1-95c0-42c8-8616-4ee70b504445
1Gi        RWX            ocs-storagecluster-cephfs     21s
```

3. Verify the metadata in the Red Hat Ceph Storage command-line interface (CLI).
   For information about how to access the Red Hat Ceph Storage CLI, see the How to access Red Hat Ceph Storage CLI in Red Hat OpenShift Data Foundation environment article.

```
$ ceph fs subvolume ls ocs-storagecluster-cephfilesystem --group_name csi
[
    {
        "name": "csi-vol-3536db13-2850-11ed-95e0-0a580a810215"
    },
    {
        "name": "csi-vol-5ea23eb0-284d-11ed-95e0-0a580a810215"
    },
    {
        "name": "csi-vol-25266061-284c-11ed-95e0-0a580a810215"
    }
]

$ ceph fs subvolume metadata ls ocs-storagecluster-cephfilesystem csi-vol-3536db13-
2850-11ed-95e0-0a580a810215 --group_name=csi --format=json

{
    "csi.ceph.com/cluster/name": "6cd7a18d-7363-4830-ad5c-f7b96927f026",
    "csi.storage.k8s.io/pv/name": "pvc-43d55ea1-95c0-42c8-8616-4ee70b504445",
    "csi.storage.k8s.io/pvc/name": "cephfs-pvc-restore",
    "csi.storage.k8s.io/pvc/namespace": "openshift-storage"
}
```

CHAPTER 4. ALERTS

4.1. SETTING UP ALERTS

For internal Mode clusters, various alerts related to the storage metrics services, storage cluster, disk devices, cluster health, cluster capacity, and so on are displayed in the Block and File, and the object dashboards. These alerts are not available for external Mode.

NOTE

It might take a few minutes for alerts to be shown in the alert panel, because only firing alerts are visible in this panel.

You can also view alerts with additional details and customize the display of Alerts in the OpenShift Container Platform.

For more information, see Managing alerts.

CHAPTER 5. REMOTE HEALTH MONITORING

OpenShift Data Foundation collects anonymized aggregated information about the health, usage, and size of clusters and reports it to Red Hat via an integrated component called Telemetry. This information allows Red Hat to improve OpenShift Data Foundation and to react to issues that impact customers more quickly.

A cluster that reports data to Red Hat via Telemetry is considered a *connected cluster*.

5.1. ABOUT TELEMETRY

Telemetry sends a carefully chosen subset of the cluster monitoring metrics to Red Hat. These metrics are sent continuously and describe:

- The size of an OpenShift Data Foundation cluster
- The health and status of OpenShift Data Foundation components
- The health and status of any upgrade being performed
- Limited usage information about OpenShift Data Foundation components and features
- Summary info about alerts reported by the cluster monitoring component

This continuous stream of data is used by Red Hat to monitor the health of clusters in real time and to react as necessary to problems that impact our customers. It also allows Red Hat to roll out OpenShift Data Foundation upgrades to customers so as to minimize service impact and continuously improve the upgrade experience.

This debugging information is available to Red Hat Support and engineering teams with the same restrictions as accessing data reported via support cases. All connected cluster information is used by Red Hat to help make OpenShift Data Foundation better and more intuitive to use. None of the information is shared with third parties.

5.2. INFORMATION COLLECTED BY TELEMETRY

Primary information collected by Telemetry includes:

- The size of the Ceph cluster in bytes : **"ceph_cluster_total_bytes"**,
- The amount of the Ceph cluster storage used in bytes : **"ceph_cluster_total_used_raw_bytes"**,
- Ceph cluster health status : **"ceph_health_status"**,
- The total count of object storage devices (OSDs) : **"job:ceph_osd_metadata:count"**,
- The total number of OpenShift Data Foundation Persistent Volumes (PVs) present in the Red Hat OpenShift Container Platform cluster : **"job:kube_pv:count"**,
- The total input/output operations per second (IOPS) (reads+writes) value for all the pools in the Ceph cluster : **"job:ceph_pools_iops:total"**,
- The total IOPS (reads+writes) value in bytes for all the pools in the Ceph cluster : **"job:ceph_pools_iops_bytes:total"**,

- The total count of the Ceph cluster versions running : **"job:ceph_versions_running:count"**
- The total number of unhealthy NooBaa buckets : **"job:noobaa_total_unhealthy_buckets:sum"**,
- The total number of NooBaa buckets : **"job:noobaa_bucket_count:sum"**,
- The total number of NooBaa objects : **"job:noobaa_total_object_count:sum"**,
- The count of NooBaa accounts : **"noobaa_accounts_num"**,
- The total usage of storage by NooBaa in bytes : **"noobaa_total_usage"**,
- The total amount of storage requested by the persistent volume claims (PVCs) from a particular storage provisioner in bytes: **"cluster:kube_persistentvolumeclaim_resource_requests_storage_bytes:provisioner:sum"**,
- The total amount of storage used by the PVCs from a particular storage provisioner in bytes: **"cluster:kubelet_volume_stats_used_bytes:provisioner:sum"**.

Telemetry does not collect identifying information such as user names, passwords, or the names or addresses of user resources.

CHAPTER 6. OPENSHIFT DATA FOUNDATION METRICS

To improve observability and make metric interpretation easier, this section outlines a reference of ceph-exporter, rook, and noobaa metrics and their functional significance.

6.1. RBD / MIRRORING

ocs_mirror_daemon_count

Mirror Daemon Count.

ocs_pool_mirroring_status

Pool Mirroring Status. 0=Disabled, 1=Enabled

ocs_rbd_client_blocklisted

State of the rbd client on a node, 0 = Unblocked, 1 = Blocked

ocs_rbd_pv_metadata

Attributes of Ceph RBD based Persistent Volume

6.2. RGW

ocs_rgw_health_status

Health Status of RGW Endpoint. 0=Connected, 1=Progressing & 2=Failure

6.3. STORAGE CLIENT/PROVIDER

ocs_storage_client_last_heartbeat

Unixtime (in sec) of last heartbeat of OCS Storage Client

ocs_storage_client_operator_version

OCS StorageClient encoded Operator Version

ocs_storage_client_storage_quota_utilization_ratio

StorageQuotaUtilizationRatio of ODF Storage Client

ocs_storage_consumer_metadata

Attributes of OCS Storage Consumers

ocs_storage_provider_operator_version

OCS StorageProvider encode Operator Version

6.4. STORAGECLUSTER

ocs_storagecluster_failure_domain_count

Count of failure domains for StorageCluster with given name and namespace

ocs_storagecluster_kms_connection_status

KMS Connection Status; 0: Connected, 1: Not Connected, 2: KMS not enabled

6.5. PROMETHEUS / HTTP HANDLER

promhttp_metric_handler_errors_total

Total number of internal errors encountered by the promhttp metric handler.

6.6. CEPH METRICS

ceph_AsyncMessenger_Worker_msgr_connection_idle_timeouts

Number of connections closed due to idleness

ceph_AsyncMessenger_Worker_msgr_connection_ready_timeouts

Number of not yet ready connections declared as dead

ceph_blk_kernel_device_bluestore_discard_op

Number of discard ops issued to kernel device

ceph_blk_kernel_device_bluestore_discard_threads

Number of discard threads running

ceph_blk_kernel_device_db_discard_op

Number of discard ops issued to kernel device

ceph_blk_kernel_device_db_discard_threads

Number of discard threads running

ceph_bluefs_alloc_db_max_lat

Max allocation latency for db device

ceph_bluefs_alloc_slow_fallback

Amount of allocations that required fallback to slow/shared device

ceph_bluefs_alloc_slow_max_lat

Max allocation latency for primary/shared device

ceph_bluefs_alloc_slow_size_fallback

Amount of allocations that required fallback to shared device's regular unit size

ceph_bluefs_alloc_unit_db

Allocation unit size (in bytes) for standalone DB device

ceph_bluefs_alloc_unit_slow

Allocation unit size (in bytes) for primary/shared device

ceph_bluefs_alloc_unit_wal

Allocation unit size (in bytes) for standalone WAL device

ceph_bluefs_alloc_wal_max_lat

Max allocation latency for wal device

ceph_bluefs_bytes_written_slow

Bytes written to WAL/SSTs at slow device

ceph_bluefs_bytes_written_sst

Bytes written to SSTs

ceph_bluefs_bytes_written_wal

Bytes written to WAL

ceph_bluefs_compact_lat_count

Average bluefs log compaction latency Count

ceph_bluefs_compact_lat_sum

Average bluefs log compaction latency Total

ceph_bluefs_compact_lock_lat_count

Average lock duration while compacting bluefs log Count

ceph_bluefs_compact_lock_lat_sum

Average lock duration while compacting bluefs log Total

ceph_bluefs_db_alloc_lat_count

Average bluefs db allocate latency Count

ceph_bluefs_db_alloc_lat_sum

Average bluefs db allocate latency Total

ceph_bluefs_db_total_bytes

Total bytes (main db device)

ceph_bluefs_db_used_bytes

Used bytes (main db device)

ceph_bluefs_flush_lat_count

Average bluefs flush latency Count

ceph_bluefs_flush_lat_sum

Average bluefs flush latency Total

ceph_bluefs_fsync_lat_count

Average bluefs fsync latency Count

ceph_bluefs_fsync_lat_sum

Average bluefs fsync latency Total

ceph_bluefs_log_bytes

Size of the metadata log

ceph_bluefs_logged_bytes

Bytes written to the metadata log

ceph_bluefs_max_bytes_db

Maximum bytes allocated from DB

ceph_bluefs_max_bytes_slow

Maximum bytes allocated from SLOW

ceph_bluefs_max_bytes_wal

Maximum bytes allocated from WAL

ceph_bluefs_num_files

File count

ceph_bluefs_read_bytes

Bytes requested in buffered read mode

ceph_bluefs_read_count

buffered read requests processed

ceph_bluefs_read_disk_bytes

Bytes read in buffered mode from disk

ceph_bluefs_read_disk_bytes_db

reads requests going to DB disk

ceph_bluefs_read_disk_bytes_slow

reads requests going to main disk

ceph_bluefs_read_disk_bytes_wal

reads requests going to WAL disk

ceph_bluefs_read_disk_count

buffered reads requests going to disk

ceph_bluefs_read_lat_count

Average bluefs read latency Count

ceph_bluefs_read_lat_sum

Average bluefs read latency Total

ceph_bluefs_read_prefetch_bytes

Bytes requested in prefetch read mode

ceph_bluefs_read_prefetch_count

prefetch read requests processed

ceph_bluefs_read_random_buffer_bytes

Bytes read from prefetch buffer in random read mode

ceph_bluefs_read_random_buffer_count

random read requests processed using prefetch buffer

ceph_bluefs_read_random_bytes

Bytes requested in random read mode

ceph_bluefs_read_random_count

random read requests processed

ceph_bluefs_read_random_disk_bytes

Bytes read from disk in random read mode

ceph_bluefs_read_random_disk_bytes_db

random reads requests going to DB disk

ceph_bluefs_read_random_disk_bytes_slow

random reads requests going to main disk

ceph_bluefs_read_random_disk_bytes_wal

random reads requests going to WAL disk

ceph_bluefs_read_random_disk_count

random reads requests going to disk

ceph_bluefs_read_random_lat_count

Average bluefs read_random latency Count

ceph_bluefs_read_random_lat_sum

Average bluefs read_random latency Total

ceph_bluefs_slow_alloc_lat_count

Average allocation latency for primary/shared device Count

ceph_bluefs_slow_alloc_lat_sum

Average allocation latency for primary/shared device Total

ceph_bluefs_slow_total_bytes

Total bytes (slow device)

ceph_bluefs_slow_used_bytes

Used bytes (slow device)

ceph_bluefs_truncate_lat_count

Average bluefs truncate latency Count

ceph_bluefs_truncate_lat_sum

Average bluefs truncate latency Total

ceph_bluefs_unlink_lat_count

Average bluefs unlink latency Count

ceph_bluefs_unlink_lat_sum

Average bluefs unlink latency Total

ceph_bluefs_wal_alloc_lat_count

Average bluefs wal allocate latency Count

ceph_bluefs_wal_alloc_lat_sum

Average bluefs wal allocate latency Total

ceph_bluefs_wal_total_bytes

Total bytes (wal device)

ceph_bluefs_wal_used_bytes

Used bytes (wal device)

ceph_bluefs_write_bytes

Bytes written

ceph_bluestore_alloc_unit

allocation unit size in bytes

ceph_bluestore_allocated

Sum for allocated bytes

ceph_bluestore_allocator_lat_count

Average bluestore allocator latency Count

ceph_bluestore_allocator_lat_sum

Average bluestore allocator latency Total

ceph_bluestore_clist_lat_count

Average collection listing latency Count

ceph_bluestore_clist_lat_sum

Average collection listing latency Total

ceph_bluestore_compress_lat_count

Average compress latency Count

ceph_bluestore_compress_lat_sum

Average compress latency Total

ceph_bluestore_compressed

Sum for stored compressed bytes

ceph_bluestore_compressed_allocated

Sum for bytes allocated for compressed data

ceph_bluestore_compressed_original

Sum for original bytes that were compressed

ceph_bluestore_csum_lat_count

Average checksum latency Count

ceph_bluestore_csum_lat_sum

Average checksum latency Total

ceph_bluestore_decompress_lat_count

Average decompress latency Count

ceph_bluestore_decompress_lat_sum

Average decompress latency Total

ceph_bluestore_fragmentation_micros

How fragmented bluestore free space is (free extents / max possible number of free extents) * 1000

ceph_bluestore_kv_commit_lat_count

Average kv_thread commit latency Count

ceph_bluestore_kv_commit_lat_sum

Average kv_thread commit latency Total

ceph_bluestore_kv_final_lat_count

Average kv_finalize thread latency Count

ceph_bluestore_kv_final_lat_sum

Average kv_finalize thread latency Total

ceph_bluestore_kv_flush_lat_count

Average kv_thread flush latency Count

ceph_bluestore_kv_flush_lat_sum

Average kv_thread flush latency Total

ceph_bluestore_kv_sync_lat_count

Average kv_sync thread latency Count

ceph_bluestore_kv_sync_lat_sum

Average kv_sync thread latency Total

ceph_bluestore_omap_get_keys_lat_count

Average omap get_keys call latency Count

ceph_bluestore_omap_get_keys_lat_sum

Average omap get_keys call latency Total

ceph_bluestore_omap_get_values_lat_count

Average omap get_values call latency Count

ceph_bluestore_omap_get_values_lat_sum

Average omap get_values call latency Total

ceph_bluestore_omap_lower_bound_lat_count

Average omap iterator lower_bound call latency Count

ceph_bluestore_omap_lower_bound_lat_sum

Average omap iterator lower_bound call latency Total

ceph_bluestore_omap_next_lat_count

Average omap iterator next call latency Count

ceph_bluestore_omap_next_lat_sum

Average omap iterator next call latency Total

ceph_bluestore_omap_seek_to_first_lat_count

Average omap iterator seek_to_first call latency Count

ceph_bluestore_omap_seek_to_first_lat_sum

Average omap iterator seek_to_first call latency Total

ceph_bluestore_omap_upper_bound_lat_count

Average omap iterator upper_bound call latency Count

ceph_bluestore_omap_upper_bound_lat_sum

Average omap iterator upper_bound call latency Total

ceph_bluestore_onode_hits

Count of onode cache lookup hits

ceph_bluestore_onode_misses

Count of onode cache lookup misses

ceph_bluestore_pricache:data_committed_bytes

total bytes committed,

ceph_bluestore_pricache:data_pri0_bytes

bytes allocated to pri0

ceph_bluestore_pricache:data_pri10_bytes

bytes allocated to pri10

ceph_bluestore_pricache:data_pri11_bytes

bytes allocated to pri11

ceph_bluestore_pricache:data_pri1_bytes

bytes allocated to pri1

ceph_bluestore_pricache:data_pri2_bytes

bytes allocated to pri2

ceph_bluestore_pricache:data_pri3_bytes

bytes allocated to pri3

ceph_bluestore_pricache:data_pri4_bytes

bytes allocated to pri4

ceph_bluestore_pricache:data_pri5_bytes

bytes allocated to pri5

ceph_bluestore_pricache:data_pri6_bytes

bytes allocated to pri6

ceph_bluestore_pricache:data_pri7_bytes

bytes allocated to pri7

ceph_bluestore_pricache:data_pri8_bytes

bytes allocated to pri8

ceph_bluestore_pricache:data_pri9_bytes

bytes allocated to pri9

ceph_bluestore_pricache:data_reserved_bytes

bytes reserved for future growth.

ceph_bluestore_pricache:kv_committed_bytes

total bytes committed,

ceph_bluestore_pricache:kv_onode_committed_bytes

total bytes committed,

ceph_bluestore_pricache:kv_onode_pri0_bytes

bytes allocated to pri0

ceph_bluestore_pricache:kv_onode_pri10_bytes

bytes allocated to pri10

ceph_bluestore_pricache:kv_onode_pri11_bytes

bytes allocated to pri11

ceph_bluestore_pricache:kv_onode_pri1_bytes

bytes allocated to pri1

ceph_bluestore_pricache:kv_onode_pri2_bytes

bytes allocated to pri2

ceph_bluestore_pricache:kv_onode_pri3_bytes

bytes allocated to pri3

ceph_bluestore_pricache:kv_onode_pri4_bytes

bytes allocated to pri4

ceph_bluestore_pricache:kv_onode_pri5_bytes

bytes allocated to pri5

ceph_bluestore_pricache:kv_onode_pri6_bytes

bytes allocated to pri6

ceph_bluestore_pricache:kv_onode_pri7_bytes

bytes allocated to pri7

ceph_bluestore_pricache:kv_onode_pri8_bytes

bytes allocated to pri8

ceph_bluestore_pricache:kv_onode_pri9_bytes

bytes allocated to pri9

ceph_bluestore_pricache:kv_onode_reserved_bytes

bytes reserved for future growth.

ceph_bluestore_pricache:kv_pri0_bytes

bytes allocated to pri0

ceph_bluestore_pricache:kv_pri10_bytes

bytes allocated to pri10

ceph_bluestore_pricache:kv_pri11_bytes

bytes allocated to pri11

ceph_bluestore_pricache:kv_pri1_bytes

bytes allocated to pri1

ceph_bluestore_pricache:kv_pri2_bytes

bytes allocated to pri2

ceph_bluestore_pricache:kv_pri3_bytes

bytes allocated to pri3

ceph_bluestore_pricache:kv_pri4_bytes

bytes allocated to pri4

ceph_bluestore_pricache:kv_pri5_bytes

bytes allocated to pri5

ceph_bluestore_pricache:kv_pri6_bytes

bytes allocated to pri6

ceph_bluestore_pricache:kv_pri7_bytes

bytes allocated to pri7

ceph_bluestore_pricache:kv_pri8_bytes

bytes allocated to pri8

ceph_bluestore_pricache:kv_pri9_bytes

bytes allocated to pri9

ceph_bluestore_pricache:kv_reserved_bytes

bytes reserved for future growth.

ceph_bluestore_pricache:meta_committed_bytes

total bytes committed,

ceph_bluestore_pricache:meta_pri0_bytes

bytes allocated to pri0

ceph_bluestore_pricache:meta_pri10_bytes

bytes allocated to pri10

ceph_bluestore_pricache:meta_pri11_bytes

bytes allocated to pri11

ceph_bluestore_pricache:meta_pri1_bytes

bytes allocated to pri1

ceph_bluestore_pricache:meta_pri2_bytes

bytes allocated to pri2

ceph_bluestore_pricache:meta_pri3_bytes

bytes allocated to pri3

ceph_bluestore_pricache:meta_pri4_bytes

bytes allocated to pri4

ceph_bluestore_pricache:meta_pri5_bytes

bytes allocated to pri5

ceph_bluestore_pricache:meta_pri6_bytes

bytes allocated to pri6

ceph_bluestore_pricache:meta_pri7_bytes

bytes allocated to pri7

ceph_bluestore_pricache:meta_pri8_bytes

bytes allocated to pri8

ceph_bluestore_pricache:meta_pri9_bytes

bytes allocated to pri9

ceph_bluestore_pricache:meta_reserved_bytes

bytes reserved for future growth.

ceph_bluestore_pricache_cache_bytes

current memory available for caches.

ceph_bluestore_pricache_heap_bytes

aggregate bytes in use by the heap

ceph_bluestore_pricache_mapped_bytes

total bytes mapped by the process

ceph_bluestore_pricache_target_bytes

target process memory usage in bytes

ceph_bluestore_pricache_unmapped_bytes

unmapped bytes that the kernel has yet to reclaim

ceph_bluestore_read_lat_count

Average read latency Count

ceph_bluestore_read_lat_sum

Average read latency Total

ceph_bluestore_read_onode_meta_lat_count

Average read onode metadata latency Count

ceph_bluestore_read_onode_meta_lat_sum

Average read onode metadata latency Total

ceph_bluestore_read_wait_aio_lat_count

Average read I/O waiting latency Count

ceph_bluestore_read_wait_aio_lat_sum

Average read I/O waiting latency Total

ceph_bluestore_reads_with_retries

Read operations that required at least one retry due to failed checksum validation

ceph_bluestore_remove_lat_count

Average removal latency Count

ceph_bluestore_remove_lat_sum

Average removal latency Total

ceph_bluestore_slow_aio_wait_count

Slow op count for aio wait

ceph_bluestore_slow_committed_kv_count

Slow op count for committed kv

ceph_bluestore_slow_read_onode_meta_count

Slow op count for read onode meta

ceph_bluestore_slow_read_wait_aio_count

Slow op count for read wait aio

ceph_bluestore_state_aio_wait_lat_count

Average aio_wait state latency Count

ceph_bluestore_state_aio_wait_lat_sum

Average aio_wait state latency Total

ceph_bluestore_state_deferred_aio_wait_lat_count

Average aio_wait state latency Count

ceph_bluestore_state_deferred_aio_wait_lat_sum

Average aio_wait state latency Total

ceph_bluestore_state_deferred_cleanup_lat_count

Average cleanup state latency Count

ceph_bluestore_state_deferred_cleanup_lat_sum

Average cleanup state latency Total

ceph_bluestore_state_deferred_queued_lat_count

Average deferred_queued state latency Count

ceph_bluestore_state_deferred_queued_lat_sum

Average deferred_queued state latency Total

ceph_bluestore_state_done_lat_count

Average done state latency Count

ceph_bluestore_state_done_lat_sum

Average done state latency Total

ceph_bluestore_state_finishing_lat_count

Average finishing state latency Count

ceph_bluestore_state_finishing_lat_sum

Average finishing state latency Total

ceph_bluestore_state_io_done_lat_count

Average io_done state latency Count

ceph_bluestore_state_io_done_lat_sum

Average io_done state latency Total

ceph_bluestore_state_kv_commiting_lat_count

Average kv_commiting state latency Count

ceph_bluestore_state_kv_commiting_lat_sum

Average kv_commiting state latency Total

ceph_bluestore_state_kv_done_lat_count

Average kv_done state latency Count

ceph_bluestore_state_kv_done_lat_sum

Average kv_done state latency Total

ceph_bluestore_state_kv_queued_lat_count

Average kv_queued state latency Count

ceph_bluestore_state_kv_queued_lat_sum

Average kv_queued state latency Total

ceph_bluestore_state_prepare_lat_count

Average prepare state latency Count

ceph_bluestore_state_prepare_lat_sum

Average prepare state latency Total

ceph_bluestore_stored

Sum for stored bytes

ceph_bluestore_truncate_lat_count

Average truncate latency Count

ceph_bluestore_truncate_lat_sum

Average truncate latency Total

ceph_bluestore_txc_commit_lat_count

Average commit latency Count

ceph_bluestore_txc_commit_lat_sum

Average commit latency Total

ceph_bluestore_txc_submit_lat_count

Average submit latency Count

ceph_bluestore_txc_submit_lat_sum

Average submit latency Total

ceph_bluestore_txc_throttle_lat_count

Average submit throttle latency Count

ceph_bluestore_txc_throttle_lat_sum

Average submit throttle latency Total

ceph_daemon_socket_up

Reports the health status of a Ceph daemon, as determined by whether it is able to respond via its admin socket (1 = healthy, 0 = unhealthy).

ceph_exporter_scrape_time

Time spent scraping and transforming perf counters to metrics

ceph_mds_cache_ireq_enqueue_scrub

Internal Request type enqueue scrub

ceph_mds_cache_ireq_exportdir

Internal Request type export dir

ceph_mds_cache_ireq_flush

Internal Request type flush

ceph_mds_cache_ireq_fragmentdir

Internal Request type fragmentdir

ceph_mds_cache_ireq_fragstats

Internal Request type frag stats

ceph_mds_cache_ireq_inodestats

Internal Request type inode stats

ceph_mds_cache_ireq_quiesce_inode

Internal Request type quiesce subvolume inode

ceph_mds_cache_ireq_quiesce_path

Internal Request type quiesce subvolume

ceph_mds_cache_num_recovering_enqueued

Files waiting for recovery

ceph_mds_cache_num_recovering_prioritized

Files waiting for recovery with elevated priority

ceph_mds_cache_num_recovering_processing

Files currently being recovered

ceph_mds_cache_num_strays

Stray dentries

ceph_mds_cache_num_strays_delayed

Stray dentries delayed

ceph_mds_cache_num_strays_enqueuing

Stray dentries enqueuing for purge

ceph_mds_cache_recovery_completed

File recoveries completed

ceph_mds_cache_recovery_started

File recoveries started

ceph_mds_cache_strays_created

Stray dentries created

ceph_mds_cache_strays_enqueued

Stray dentries enqueued for purge

ceph_mds_cache_strays_migrated

Stray dentries migrated

ceph_mds_cache_strays_reintegrated

Stray dentries reintegrated

ceph_mds_caps

Capabilities

ceph_mds_ceph_cap_op_flush_ack

caps truncate notify

ceph_mds_ceph_cap_op_flushsnap_ack

caps truncate notify

ceph_mds_ceph_cap_op_grant

Grant caps

ceph_mds_ceph_cap_op_revoke

Revoke caps

ceph_mds_ceph_cap_op_trunc

caps truncate notify

ceph_mds_client_metrics_num_clients

Numer of client sessions

ceph_mds_dir_commit

Directory commit

ceph_mds_dir_fetch_complete

Fetch complete dirfrag

ceph_mds_dir_fetch_keys

Fetch keys from dirfrag

ceph_mds_dir_merge

Directory merge

ceph_mds_dir_split

Directory split

ceph_mds_exported_inodes

Exported inodes

ceph_mds_forward

Forwarding request

ceph_mds_handle_client_cap_release

Client cap release msg

ceph_mds_handle_client_caps

Client caps msg

ceph_mds_handle_client_caps_dirty

Client dirty caps msg

ceph_mds_handle_inode_file_caps

Inter mds caps msg

ceph_mds_imported_inodes

Imported inodes

ceph_mds_inodes

Inodes

ceph_mds_inodes_expired

Inodes expired

ceph_mds_inodes_pinned

Inodes pinned

ceph_mds_inodes_with_caps

Inodes with capabilities

ceph_mds_load_cent

Load per cent

ceph_mds_log_ev

Events

ceph_mds_log_evadd

Events submitted

ceph_mds_log_evex

Total expired events

ceph_mds_log_evexd

Current expired events

ceph_mds_log_evexg

Expiring events

ceph_mds_log_evlrg

Large events

ceph_mds_log_evtrm

Trimmed events

ceph_mds_log_jlat_count

Journaler flush latency Count

ceph_mds_log_jlat_sum

Journaler flush latency Total

ceph_mds_log_replayed

Events replayed

ceph_mds_log_seg

Segments

ceph_mds_log_segadd

Segments added

ceph_mds_log_segex

Total expired segments

ceph_mds_log_segexd

Current expired segments

ceph_mds_log_segexg

Expiring segments

ceph_mds_log_segmjr

Major Segments

ceph_mds_log_segtrm

Trimmed segments

ceph_mds_mem_cap

Capabilities

ceph_mds_mem_cap_minus

Capabilities removed

ceph_mds_mem_cap_plus

Capabilities added

ceph_mds_mem_dir

Directories

ceph_mds_mem_dir_minus

Directories closed

ceph_mds_mem_dir_plus

Directories opened

ceph_mds_mem_dn

Dentries

ceph_mds_mem_dn_minus

Dentries closed

ceph_mds_mem_dn_plus

Dentries opened

ceph_mds_mem_heap

Heap size

ceph_mds_mem_ino

Inodes

ceph_mds_mem_ino_minus

Inodes closed

ceph_mds_mem_ino_plus

Inodes opened

ceph_mds_mem_rss

RSS

ceph_mds_openino_dir_fetch

OpenIno incomplete directory fetchings

ceph_mds_process_request_cap_release

Process request cap release

ceph_mds_reply_latency_count

Reply latency Count

ceph_mds_reply_latency_sum

Reply latency Total

ceph_mds_request

Requests

ceph_mds_root_rbytes

root inode rbytes

ceph_mds_root_rfiles

root inode rfiles

ceph_mds_root_rsnaps

root inode rsnaps

ceph_mds_server_cap_acquisition_throttle

Cap acquisition throttle counter

ceph_mds_server_cap_revoke_eviction

Cap Revoke Client Eviction

ceph_mds_server_handle_client_request

Client requests

ceph_mds_server_handle_client_session

Client session messages

ceph_mds_server_handle_peer_request

Peer requests

ceph_mds_server_req_blockdiff_latency_count

Request type file blockdiff latency Count

ceph_mds_server_req_blockdiff_latency_sum

Request type file blockdiff latency Total

ceph_mds_server_req_create_latency_count

Request type create latency Count

ceph_mds_server_req_create_latency_sum

Request type create latency Total

ceph_mds_server_req_getattr_latency_count

Request type get attribute latency Count

ceph_mds_server_req_getattr_latency_sum

Request type get attribute latency Total

ceph_mds_server_req_getfilelock_latency_count

Request type get file lock latency Count

ceph_mds_server_req_getfilelock_latency_sum

Request type get file lock latency Total

ceph_mds_server_req_getvxattr_latency_count

Request type get virtual extended attribute latency Count

ceph_mds_server_req_getvxattr_latency_sum

Request type get virtual extended attribute latency Total

ceph_mds_server_req_link_latency_count

Request type link latency Count

ceph_mds_server_req_link_latency_sum

Request type link latency Total

ceph_mds_server_req_lookup_latency_count

Request type lookup latency Count

ceph_mds_server_req_lookup_latency_sum

Request type lookup latency Total

ceph_mds_server_req_lookuphash_latency_count

Request type lookup hash of inode latency Count

ceph_mds_server_req_lookuphash_latency_sum

Request type lookup hash of inode latency Total

ceph_mds_server_req_lookupino_latency_count

Request type lookup inode latency Count

ceph_mds_server_req_lookupino_latency_sum

Request type lookup inode latency Total

ceph_mds_server_req_lookupname_latency_count

Request type lookup name latency Count

ceph_mds_server_req_lookupname_latency_sum

Request type lookup name latency Total

ceph_mds_server_req_lookupparent_latency_count

Request type lookup parent latency Count

ceph_mds_server_req_lookupparent_latency_sum

Request type lookup parent latency Total

ceph_mds_server_req_lookupsnap_latency_count

Request type lookup snapshot latency Count

ceph_mds_server_req_lookupsnap_latency_sum

Request type lookup snapshot latency Total

ceph_mds_server_req_lssnap_latency_count

Request type list snapshot latency Count

ceph_mds_server_req_lssnap_latency_sum

Request type list snapshot latency Total

ceph_mds_server_req_mkdir_latency_count

Request type make directory latency Count

ceph_mds_server_req_mkdir_latency_sum

Request type make directory latency Total

ceph_mds_server_req_mknod_latency_count

Request type make node latency Count

ceph_mds_server_req_mknod_latency_sum

Request type make node latency Total

ceph_mds_server_req_mksnap_latency_count

Request type make snapshot latency Count

ceph_mds_server_req_mksnap_latency_sum

Request type make snapshot latency Total

ceph_mds_server_req_open_latency_count

Request type open latency Count

ceph_mds_server_req_open_latency_sum

Request type open latency Total

ceph_mds_server_req_readdir_latency_count

Request type read directory latency Count

ceph_mds_server_req_readdir_latency_sum

Request type read directory latency Total

ceph_mds_server_req_rename_latency_count

Request type rename latency Count

ceph_mds_server_req_rename_latency_sum

Request type rename latency Total

ceph_mds_server_req_renamesnap_latency_count

Request type rename snapshot latency Count

ceph_mds_server_req_renamesnap_latency_sum

Request type rename snapshot latency Total

ceph_mds_server_req_rmdir_latency_count

Request type remove directory latency Count

ceph_mds_server_req_rmdir_latency_sum

Request type remove directory latency Total

ceph_mds_server_req_rmsnap_latency_count

Request type remove snapshot latency Count

ceph_mds_server_req_rmsnap_latency_sum

Request type remove snapshot latency Total

ceph_mds_server_req_rmxattr_latency_count

Request type remove extended attribute latency Count

ceph_mds_server_req_rmxattr_latency_sum

Request type remove extended attribute latency Total

ceph_mds_server_req_setattr_latency_count

Request type set attribute latency Count

ceph_mds_server_req_setattr_latency_sum

Request type set attribute latency Total

ceph_mds_server_req_setdirlayout_latency_count

Request type set directory layout latency Count

ceph_mds_server_req_setdirlayout_latency_sum

Request type set directory layout latency Total

ceph_mds_server_req_setfilelock_latency_count

Request type set file lock latency Count

ceph_mds_server_req_setfilelock_latency_sum

Request type set file lock latency Total

ceph_mds_server_req_setlayout_latency_count

Request type set file layout latency Count

ceph_mds_server_req_setlayout_latency_sum

Request type set file layout latency Total

ceph_mds_server_req_setxattr_latency_count

Request type set extended attribute latency Count

ceph_mds_server_req_setxattr_latency_sum

Request type set extended attribute latency Total

ceph_mds_server_req_snapdiff_latency_count

Request type snapshot difference latency Count

ceph_mds_server_req_snapdiff_latency_sum

Request type snapshot difference latency Total

ceph_mds_server_req_symlink_latency_count

Request type symbolic link latency Count

ceph_mds_server_req_symlink_latency_sum

Request type symbolic link latency Total

ceph_mds_server_req_unlink_latency_count

Request type unlink latency Count

ceph_mds_server_req_unlink_latency_sum

Request type unlink latency Total

ceph_mds_sessions_average_load

Average Load

ceph_mds_sessions_avg_session_uptime

Average session uptime

ceph_mds_sessions_mdthresh_evicted

Sessions evicted on reaching metadata threshold

ceph_mds_sessions_session_add

Sessions added

ceph_mds_sessions_session_count

Session count

ceph_mds_sessions_session_remove

Sessions removed

ceph_mds_sessions_sessions_open

Sessions currently open

ceph_mds_sessions_sessions_stale

Sessions currently stale

ceph_mds_sessions_total_load

Total Load

ceph_mds_slow_reply

Slow replies

ceph_mds_subtrees

Subtrees

ceph_mon_election_call

Elections started

ceph_mon_election_lose

Elections lost

ceph_mon_election_win

Elections won

ceph_mon_num_elections

Elections participated in

ceph_mon_num_sessions

Open sessions

ceph_mon_session_add

Created sessions

ceph_mon_session_rm

Removed sessions

ceph_mon_session_trim

Trimmed sessions

ceph_objecter_op_active

Operations active

ceph_objecter_op_r

Read operations

ceph_objecter_op_rmw

Read-modify-write operations

ceph_objecter_op_w

Write operations

ceph_osd_numpg

Placement groups

ceph_osd_numpg_removing

Placement groups queued for local deletion

ceph_osd_op

Client operations

ceph_osd_op_before_queue_op_lat_count

Latency of IO before calling queue(before really queue into ShardedOpWq) Count

ceph_osd_op_before_queue_op_lat_sum

Latency of IO before calling queue(before really queue into ShardedOpWq) Total

ceph_osd_op_delayed_degraded

Count of ops delayed due to target object being degraded

ceph_osd_op_delayed_unreadable

Count of ops delayed due to target object being unreadable

ceph_osd_op_in_bytes

Client operations total write size

ceph_osd_op_latency_count

Latency of client operations (including queue time) Count

ceph_osd_op_latency_sum

Latency of client operations (including queue time) Total

ceph_osd_op_out_bytes

Client operations total read size

ceph_osd_op_prepare_latency_count

Latency of client operations (excluding queue time and wait for finished) Count

ceph_osd_op_prepare_latency_sum

Latency of client operations (excluding queue time and wait for finished) Total

ceph_osd_op_process_latency_count

Latency of client operations (excluding queue time) Count

ceph_osd_op_process_latency_sum

Latency of client operations (excluding queue time) Total

ceph_osd_op_r

Client read operations

ceph_osd_op_r_latency_count

Latency of read operation (including queue time) Count

ceph_osd_op_r_latency_sum

Latency of read operation (including queue time) Total

ceph_osd_op_r_out_bytes

Client data read

ceph_osd_op_r_prepare_latency_count

Latency of read operations (excluding queue time and wait for finished) Count

ceph_osd_op_r_prepare_latency_sum

Latency of read operations (excluding queue time and wait for finished) Total

ceph_osd_op_r_process_latency_count

Latency of read operation (excluding queue time) Count

ceph_osd_op_r_process_latency_sum

Latency of read operation (excluding queue time) Total

ceph_osd_op_rw

Client read-modify-write operations

ceph_osd_op_rw_in_bytes

Client read-modify-write operations write in

ceph_osd_op_rw_latency_count

Latency of read-modify-write operation (including queue time) Count

ceph_osd_op_rw_latency_sum

Latency of read-modify-write operation (including queue time) Total

ceph_osd_op_rw_out_bytes

Client read-modify-write operations read out

ceph_osd_op_rw_prepare_latency_count

Latency of read-modify-write operations (excluding queue time and wait for finished) Count

ceph_osd_op_rw_prepare_latency_sum

Latency of read-modify-write operations (excluding queue time and wait for finished) Total

ceph_osd_op_rw_process_latency_count

Latency of read-modify-write operation (excluding queue time) Count

ceph_osd_op_rw_process_latency_sum

Latency of read-modify-write operation (excluding queue time) Total

ceph_osd_op_w

Client write operations

ceph_osd_op_w_in_bytes

Client data written

ceph_osd_op_w_latency_count

Latency of write operation (including queue time) Count

ceph_osd_op_w_latency_sum

Latency of write operation (including queue time) Total

ceph_osd_op_w_prepare_latency_count

Latency of write operations (excluding queue time and wait for finished) Count

ceph_osd_op_w_prepare_latency_sum

Latency of write operations (excluding queue time and wait for finished) Total

ceph_osd_op_w_process_latency_count

Latency of write operation (excluding queue time) Count

ceph_osd_op_w_process_latency_sum

Latency of write operation (excluding queue time) Total

ceph_osd_op_wip

Replication operations currently being processed (primary)

ceph_osd_recovery_bytes

recovery bytes

ceph_osd_recovery_ops

Started recovery operations

ceph_osd_scrub_dp_ec_chunk_busy

chunk busy during scrubs

ceph_osd_scrub_dp_ec_chunk_selected

chunk selection during scrubs

ceph_osd_scrub_dp_ec_failed_reservations_elapsed_count

time for scrub reservation to fail Count

ceph_osd_scrub_dp_ec_failed_reservations_elapsed_sum

time for scrub reservation to fail Total

ceph_osd_scrub_dp_ec_failed_scrubs

failed scrubs count

ceph_osd_scrub_dp_ec_failed_scrubs_elapsed_count

time to scrub failure Count

ceph_osd_scrub_dp_ec_failed_scrubs_elapsed_sum

time to scrub failure Total

ceph_osd_scrub_dp_ec_locked_object

waiting on locked object events

ceph_osd_scrub_dp_ec_num_scrubs_past_reservation

scrubs count

ceph_osd_scrub_dp_ec_num_scrubs_started

scrubs attempted count

ceph_osd_scrub_dp_ec_preemptions

preemptions on scrubs

ceph_osd_scrub_dp_ec_replicas_in_reservation

number of replicas in reservation

ceph_osd_scrub_dp_ec_reservation_process_aborted

scrub reservation was aborted

ceph_osd_scrub_dp_ec_reservation_process_failure

scrub reservation failed due to replica denial

ceph_osd_scrub_dp_ec_reservation_process_skipped

scrub reservation skipped for high priority scrub

ceph_osd_scrub_dp_ec_scrub_reservations_completed

successfully completed reservation processes

ceph_osd_scrub_dp_ec_successful_reservations_elapsed_count

time to scrub reservation completion Count

ceph_osd_scrub_dp_ec_successful_reservations_elapsed_sum

time to scrub reservation completion Total

ceph_osd_scrub_dp_ec_successful_scrubs

successful scrubs count

ceph_osd_scrub_dp_ec_successful_scrubs_elapsed_count

time to scrub completion Count

ceph_osd_scrub_dp_ec_successful_scrubs_elapsed_sum

time to scrub completion Total

ceph_osd_scrub_dp_ec_write_blocked_by_scrub

write blocked by scrub

ceph_osd_scrub_dp_repl_chunk_busy

chunk busy during scrubs

ceph_osd_scrub_dp_repl_chunk_selected

chunk selection during scrubs

ceph_osd_scrub_dp_repl_failed_reservations_elapsed_count

time for scrub reservation to fail Count

ceph_osd_scrub_dp_repl_failed_reservations_elapsed_sum

time for scrub reservation to fail Total

ceph_osd_scrub_dp_repl_failed_scrubs

failed scrubs count

ceph_osd_scrub_dp_repl_failed_scrubs_elapsed_count

time to scrub failure Count

ceph_osd_scrub_dp_repl_failed_scrubs_elapsed_sum

time to scrub failure Total

ceph_osd_scrub_dp_repl_locked_object

waiting on locked object events

ceph_osd_scrub_dp_repl_num_scrubs_past_reservation

scrubs count

ceph_osd_scrub_dp_repl_num_scrubs_started

scrubs attempted count

ceph_osd_scrub_dp_repl_preemptions

preemptions on scrubs

ceph_osd_scrub_dp_repl_replicas_in_reservation

number of replicas in reservation

ceph_osd_scrub_dp_repl_reservation_process_aborted

scrub reservation was aborted

ceph_osd_scrub_dp_repl_reservation_process_failure

scrub reservation failed due to replica denial

ceph_osd_scrub_dp_repl_reservation_process_skipped

scrub reservation skipped for high priority scrub

ceph_osd_scrub_dp_repl_scrub_reservations_completed

successfully completed reservation processes

ceph_osd_scrub_dp_repl_successful_reservations_elapsed_count

time to scrub reservation completion Count

ceph_osd_scrub_dp_repl_successful_reservations_elapsed_sum

time to scrub reservation completion Total

ceph_osd_scrub_dp_repl_successful_scrubs

successful scrubs count

ceph_osd_scrub_dp_repl_successful_scrubs_elapsed_count

time to scrub completion Count

ceph_osd_scrub_dp_repl_successful_scrubs_elapsed_sum

time to scrub completion Total

ceph_osd_scrub_dp_repl_write_blocked_by_scrub

write blocked by scrub

ceph_osd_scrub_sh_ec_chunk_busy

chunk busy during scrubs

ceph_osd_scrub_sh_ec_chunk_selected

chunk selection during scrubs

ceph_osd_scrub_sh_ec_failed_reservations_elapsed_count

time for scrub reservation to fail Count

ceph_osd_scrub_sh_ec_failed_reservations_elapsed_sum

time for scrub reservation to fail Total

ceph_osd_scrub_sh_ec_failed_scrubs

failed scrubs count

ceph_osd_scrub_sh_ec_failed_scrubs_elapsed_count

time to scrub failure Count

ceph_osd_scrub_sh_ec_failed_scrubs_elapsed_sum

time to scrub failure Total

ceph_osd_scrub_sh_ec_locked_object

waiting on locked object events

ceph_osd_scrub_sh_ec_num_scrubs_past_reservation

scrubs count

ceph_osd_scrub_sh_ec_num_scrubs_started

scrubs attempted count

ceph_osd_scrub_sh_ec_preemptions

preemptions on scrubs

ceph_osd_scrub_sh_ec_replicas_in_reservation

number of replicas in reservation

ceph_osd_scrub_sh_ec_reservation_process_aborted

scrub reservation was aborted

ceph_osd_scrub_sh_ec_reservation_process_failure

scrub reservation failed due to replica denial

ceph_osd_scrub_sh_ec_reservation_process_skipped

scrub reservation skipped for high priority scrub

ceph_osd_scrub_sh_ec_scrub_reservations_completed

successfully completed reservation processes

ceph_osd_scrub_sh_ec_successful_reservations_elapsed_count

time to scrub reservation completion Count

ceph_osd_scrub_sh_ec_successful_reservations_elapsed_sum

time to scrub reservation completion Total

ceph_osd_scrub_sh_ec_successful_scrubs

successful scrubs count

ceph_osd_scrub_sh_ec_successful_scrubs_elapsed_count

time to scrub completion Count

ceph_osd_scrub_sh_ec_successful_scrubs_elapsed_sum

time to scrub completion Total

ceph_osd_scrub_sh_ec_write_blocked_by_scrub

write blocked by scrub

ceph_osd_scrub_sh_repl_chunk_busy

chunk busy during scrubs

ceph_osd_scrub_sh_repl_chunk_selected

chunk selection during scrubs

ceph_osd_scrub_sh_repl_failed_reservations_elapsed_count

time for scrub reservation to fail Count

ceph_osd_scrub_sh_repl_failed_reservations_elapsed_sum

time for scrub reservation to fail Total

ceph_osd_scrub_sh_repl_failed_scrubs

failed scrubs count

ceph_osd_scrub_sh_repl_failed_scrubs_elapsed_count

time to scrub failure Count

ceph_osd_scrub_sh_repl_failed_scrubs_elapsed_sum

time to scrub failure Total

ceph_osd_scrub_sh_repl_locked_object

waiting on locked object events

ceph_osd_scrub_sh_repl_num_scrubs_past_reservation

scrubs count

ceph_osd_scrub_sh_repl_num_scrubs_started

scrubs attempted count

ceph_osd_scrub_sh_repl_preemptions

preemptions on scrubs

ceph_osd_scrub_sh_repl_replicas_in_reservation

number of replicas in reservation

ceph_osd_scrub_sh_repl_reservation_process_aborted

scrub reservation was aborted

ceph_osd_scrub_sh_repl_reservation_process_failure

scrub reservation failed due to replica denial

ceph_osd_scrub_sh_repl_reservation_process_skipped

scrub reservation skipped for high priority scrub

ceph_osd_scrub_sh_repl_scrub_reservations_completed

successfully completed reservation processes

ceph_osd_scrub_sh_repl_successful_reservations_elapsed_count

time to scrub reservation completion Count

ceph_osd_scrub_sh_repl_successful_reservations_elapsed_sum

time to scrub reservation completion Total

ceph_osd_scrub_sh_repl_successful_scrubs

successful scrubs count

ceph_osd_scrub_sh_repl_successful_scrubs_elapsed_count

time to scrub completion Count

ceph_osd_scrub_sh_repl_successful_scrubs_elapsed_sum

time to scrub completion Total

ceph_osd_scrub_sh_repl_write_blocked_by_scrub

write blocked by scrub

ceph_osd_stat_bytes

OSD size

ceph_osd_stat_bytes_used

Used space

ceph_paxos_accept_timeout

Accept timeouts

ceph_paxos_begin

Started and handled begins

ceph_paxos_begin_bytes_count

Data in transaction on begin Count

ceph_paxos_begin_bytes_sum

Data in transaction on begin Total

ceph_paxos_begin_keys_count

Keys in transaction on begin Count

ceph_paxos_begin_keys_sum

Keys in transaction on begin Total

ceph_paxos_begin_latency_count

Latency of begin operation Count

ceph_paxos_begin_latency_sum

Latency of begin operation Total

ceph_paxos_collect

Peon collects

ceph_paxos_collect_bytes_count

Data in transaction on peon collect Count

ceph_paxos_collect_bytes_sum

Data in transaction on peon collect Total

ceph_paxos_collect_keys_count

Keys in transaction on peon collect Count

ceph_paxos_collect_keys_sum

Keys in transaction on peon collect Total

ceph_paxos_collect_latency_count

Peon collect latency Count

ceph_paxos_collect_latency_sum

Peon collect latency Total

ceph_paxos_collect_timeout

Collect timeouts

ceph_paxos_collect_uncommitted

Uncommitted values in started and handled collects

ceph_paxos_commit

Commits

ceph_paxos_commit_bytes_count

Data in transaction on commit Count

ceph_paxos_commit_bytes_sum

Data in transaction on commit Total

ceph_paxos_commit_keys_count

Keys in transaction on commit Count

ceph_paxos_commit_keys_sum

Keys in transaction on commit Total

ceph_paxos_commit_latency_count

Commit latency Count

ceph_paxos_commit_latency_sum

Commit latency Total

ceph_paxos_lease_ack_timeout

Lease acknowledgement timeouts

ceph_paxos_lease_timeout

Lease timeouts

ceph_paxos_new_pn

New proposal number queries

ceph_paxos_new_pn_latency_count

New proposal number getting latency Count

ceph_paxos_new_pn_latency_sum

New proposal number getting latency Total

ceph_paxos_refresh

Refreshes

ceph_paxos_refresh_latency_count

Refresh latency Count

ceph_paxos_refresh_latency_sum

Refresh latency Total

ceph_paxos_restart

Restarts

ceph_paxos_share_state

Sharings of state

ceph_paxos_share_state_bytes_count

Data in shared state Count

ceph_paxos_share_state_bytes_sum

Data in shared state Total

ceph_paxos_share_state_keys_count

Keys in shared state Count

ceph_paxos_share_state_keys_sum

Keys in shared state Total

ceph_paxos_start_leader

Starts in leader role

ceph_paxos_start_peon

Starts in peon role

ceph_paxos_store_state

Store a shared state on disk

ceph_paxos_store_state_bytes_count

Data in transaction in stored state Count

ceph_paxos_store_state_bytes_sum

Data in transaction in stored state Total

ceph_paxos_store_state_keys_count

Keys in transaction in stored state Count

ceph_paxos_store_state_keys_sum

Keys in transaction in stored state Total

ceph_paxos_store_state_latency_count

Storing state latency Count

ceph_paxos_store_state_latency_sum

Storing state latency Total

ceph_prioritycache:full_committed_bytes

total bytes committed,

ceph_prioritycache:full_pri0_bytes

bytes allocated to pri0

ceph_prioritycache:full_pri10_bytes

bytes allocated to pri10

ceph_prioritycache:full_pri11_bytes

bytes allocated to pri11

ceph_prioritycache:full_pri1_bytes

bytes allocated to pri1

ceph_prioritycache:full_pri2_bytes

bytes allocated to pri2

ceph_prioritycache:full_pri3_bytes

bytes allocated to pri3

ceph_prioritycache:full_pri4_bytes

bytes allocated to pri4

ceph_prioritycache:full_pri5_bytes

bytes allocated to pri5

ceph_prioritycache:full_pri6_bytes

bytes allocated to pri6

ceph_prioritycache:full_pri7_bytes

bytes allocated to pri7

ceph_prioritycache:full_pri8_bytes

bytes allocated to pri8

ceph_prioritycache:full_pri9_bytes

bytes allocated to pri9

ceph_prioritycache:full_reserved_bytes

bytes reserved for future growth.

ceph_prioritycache:inc_committed_bytes

total bytes committed,

ceph_prioritycache:inc_pri0_bytes

bytes allocated to pri0

ceph_prioritycache:inc_pri10_bytes

bytes allocated to pri10

ceph_prioritycache:inc_pri11_bytes

bytes allocated to pri11

ceph_prioritycache:inc_pri1_bytes

bytes allocated to pri1

ceph_prioritycache:inc_pri2_bytes

bytes allocated to pri2

ceph_prioritycache:inc_pri3_bytes

bytes allocated to pri3

ceph_prioritycache:inc_pri4_bytes

bytes allocated to pri4

ceph_prioritycache:inc_pri5_bytes

bytes allocated to pri5

ceph_prioritycache:inc_pri6_bytes

bytes allocated to pri6

ceph_prioritycache:inc_pri7_bytes

bytes allocated to pri7

ceph_prioritycache:inc_pri8_bytes

bytes allocated to pri8

ceph_prioritycache:inc_pri9_bytes

bytes allocated to pri9

ceph_prioritycache:inc_reserved_bytes

bytes reserved for future growth.

ceph_prioritycache:kv_committed_bytes

total bytes committed,

ceph_prioritycache:kv_pri0_bytes

bytes allocated to pri0

ceph_prioritycache:kv_pri10_bytes

bytes allocated to pri10

ceph_prioritycache:kv_pri11_bytes

bytes allocated to pri11

ceph_prioritycache:kv_pri1_bytes

bytes allocated to pri1

ceph_prioritycache:kv_pri2_bytes

bytes allocated to pri2

ceph_prioritycache:kv_pri3_bytes

bytes allocated to pri3

ceph_prioritycache:kv_pri4_bytes

bytes allocated to pri4

ceph_prioritycache:kv_pri5_bytes

bytes allocated to pri5

ceph_prioritycache:kv_pri6_bytes

bytes allocated to pri6

ceph_prioritycache:kv_pri7_bytes

bytes allocated to pri7

ceph_prioritycache:kv_pri8_bytes

bytes allocated to pri8

ceph_prioritycache:kv_pri9_bytes

bytes allocated to pri9

ceph_prioritycache:kv_reserved_bytes

bytes reserved for future growth.

ceph_prioritycache_cache_bytes

current memory available for caches.

ceph_prioritycache_heap_bytes

aggregate bytes in use by the heap

ceph_prioritycache_mapped_bytes

total bytes mapped by the process

ceph_prioritycache_target_bytes

target process memory usage in bytes

ceph_prioritycache_unmapped_bytes

unmapped bytes that the kernel has yet to reclaim

ceph_purge_queue_pq_executed

Purge queue tasks executed

ceph_purge_queue_pq_executed_ops

Purge queue ops executed

ceph_purge_queue_pq_executing

Purge queue tasks in flight

ceph_purge_queue_pq_executing_high_water

Maximum number of executing file purges

ceph_purge_queue_pq_executing_ops

Purge queue ops in flight

ceph_purge_queue_pq_executing_ops_high_water

Maximum number of executing file purge ops

ceph_purge_queue_pq_item_in_journal

Purge item left in journal

ceph_rocksdb_compact

Compactions

ceph_rocksdb_compact_completed

Completed compactions

ceph_rocksdb_compact_lasted

Last completed compaction duration

ceph_rocksdb_compact_queue_len

Length of compaction queue

ceph_rocksdb_compact_queue_merge

Mergings of ranges in compaction queue

ceph_rocksdb_compact_running

Running compactions

ceph_rocksdb_get_latency_count

Get latency Count

ceph_rocksdb_get_latency_sum

Get latency Total

ceph_rocksdb_rocksdb_write_delay_time_count

Rocksdb write delay time Count

ceph_rocksdb_rocksdb_write_delay_time_sum

Rocksdb write delay time Total

ceph_rocksdb_rocksdb_write_memtable_time_count

Rocksdb write memtable time Count

ceph_rocksdb_rocksdb_write_memtable_time_sum

Rocksdb write memtable time Total

ceph_rocksdb_rocksdb_write_pre_and_post_time_count

total time spent on writing a record, excluding write process Count

ceph_rocksdb_rocksdb_write_pre_and_post_time_sum

total time spent on writing a record, excluding write process Total

ceph_rocksdb_rocksdb_write_wal_time_count

Rocksdb write wal time Count

ceph_rocksdb_rocksdb_write_wal_time_sum

Rocksdb write wal time Total

ceph_rocksdb_submit_latency_count

Submit Latency Count

ceph_rocksdb_submit_latency_sum

Submit Latency Total

ceph_rocksdb_submit_sync_latency_count

Submit Sync Latency Count

ceph_rocksdb_submit_sync_latency_sum

Submit Sync Latency Total

6.7. NOOBAA

NooBaa_Endpoint_process_cpu_user_seconds_total

Total user CPU time spent in seconds.

NooBaa_Endpoint_process_cpu_system_seconds_total

Total system CPU time spent in seconds.

NooBaa_Endpoint_process_cpu_seconds_total

Total user and system CPU time spent in seconds.

NooBaa_Endpoint_process_start_time_seconds

Start time of the process since unix epoch in seconds.

NooBaa_Endpoint_process_resident_memory_bytes

Resident memory size in bytes.

NooBaa_Endpoint_process_virtual_memory_bytes

Virtual memory size in bytes.

NooBaa_Endpoint_process_heap_bytes

Process heap size in bytes.

NooBaa_Endpoint_process_open_fds

Number of open file descriptors.

NooBaa_Endpoint_process_max_fds

Maximum number of open file descriptors.

NooBaa_Endpoint_nodejs_eventloop_lag_seconds

Lag of event loop in seconds.

NooBaa_Endpoint_nodejs_eventloop_lag_min_seconds

The minimum recorded event loop delay.

NooBaa_Endpoint_nodejs_eventloop_lag_max_seconds

The maximum recorded event loop delay.

NooBaa_Endpoint_nodejs_eventloop_lag_mean_seconds

The mean of the recorded event loop delays.

NooBaa_Endpoint_nodejs_eventloop_lag_stddev_seconds

The standard deviation of the recorded event loop delays.

NooBaa_Endpoint_nodejs_eventloop_lag_p50_seconds

The 50th percentile of the recorded event loop delays.

NooBaa_Endpoint_nodejs_eventloop_lag_p90_seconds

The 90th percentile of the recorded event loop delays.

NooBaa_Endpoint_nodejs_eventloop_lag_p99_seconds

The 99th percentile of the recorded event loop delays.

NooBaa_Endpoint_nodejs_active_resources

Number of active resources that are currently keeping the event loop alive, grouped by async resource type.

NooBaa_Endpoint_nodejs_active_resources_total

Total number of active resources.

NooBaa_Endpoint_nodejs_active_handles

Number of active libuv handles grouped by handle type. Every handle type is C++ class name.

NooBaa_Endpoint_nodejs_active_handles_total

Total number of active handles.

NooBaa_Endpoint_nodejs_active_requests

Number of active libuv requests grouped by request type. Every request type is C++ class name.

NooBaa_Endpoint_nodejs_active_requests_total

Total number of active requests.

NooBaa_Endpoint_nodejs_heap_size_total_bytes

Process heap size from Node.js in bytes.

NooBaa_Endpoint_nodejs_heap_size_used_bytes

Process heap size used from Node.js in bytes.

NooBaa_Endpoint_nodejs_external_memory_bytes

Node.js external memory size in bytes.

NooBaa_Endpoint_nodejs_heap_space_size_total_bytes

Process heap space size total from Node.js in bytes.

NooBaa_Endpoint_nodejs_heap_space_size_used_bytes

Process heap space size used from Node.js in bytes.

NooBaa_Endpoint_nodejs_heap_space_size_available_bytes

Process heap space size available from Node.js in bytes.

NooBaa_Endpoint_nodejs_version_info

Node.js version info.

NooBaa_Endpoint_nodejs_gc_duration_seconds

Garbage collection duration by kind, one of major, minor, incremental or weakcb.

NooBaa_Endpoint_hub_read_bytes

hub read bytes in namespace cache bucket

NooBaa_Endpoint_hub_write_bytes

hub write bytes in namespace cache bucket

NooBaa_Endpoint_cache_read_bytes

Cache read bytes in namespace cache bucket

NooBaa_Endpoint_cache_write_bytes

Cache write bytes in namespace cache bucket

NooBaa_Endpoint_cache_object_read_count

Counter on entire object reads in namespace cache bucket

NooBaa_Endpoint_cache_object_read_miss_count

Counter on entire object read miss in namespace cache bucket

NooBaa_Endpoint_cache_object_read_hit_count

Counter on entire object read hit in namespace cache bucket

NooBaa_Endpoint_cache_range_read_count

Counter on range reads in namespace cache bucket

NooBaa_Endpoint_cache_range_read_miss_count

Counter on range read miss in namespace cache bucket

NooBaa_Endpoint_cache_range_read_hit_count

Counter on range read hit in namespace cache bucket

NooBaa_Endpoint_hub_read_latency

hub read latency in namespace cache bucket

NooBaa_Endpoint_hub_write_latency

hub write latency in namespace cache bucket

NooBaa_Endpoint_cache_read_latency

Cache read latency in namespace cache bucket

NooBaa_Endpoint_cache_write_latency

Cache write latency in namespace cache bucket

NooBaa_Endpoint_semaphore_waiting_value

Namespace semaphore waiting value

NooBaa_Endpoint_semaphore_waiting_time

Namespace semaphore waiting time

NooBaa_Endpoint_semaphore_waiting_queue

Namespace semaphore waiting queue size

NooBaa_Endpoint_semaphore_value

Namespace semaphore value

NooBaa_Endpoint_fork_counter

Counter on number of fork hit

NooBaa_WebServer_process_cpu_user_seconds_total

Total user CPU time spent in seconds.

NooBaa_WebServer_process_cpu_system_seconds_total

Total system CPU time spent in seconds.

NooBaa_WebServer_process_cpu_seconds_total

Total user and system CPU time spent in seconds.

NooBaa_WebServer_process_start_time_seconds

Start time of the process since unix epoch in seconds.

NooBaa_WebServer_process_resident_memory_bytes

Resident memory size in bytes.

NooBaa_WebServer_process_virtual_memory_bytes

Virtual memory size in bytes.

NooBaa_WebServer_process_heap_bytes

Process heap size in bytes.

NooBaa_WebServer_process_open_fds

Number of open file descriptors.

NooBaa_WebServer_process_max_fds

Maximum number of open file descriptors.

NooBaa_WebServer_nodejs_eventloop_lag_seconds

Lag of event loop in seconds.

NooBaa_WebServer_nodejs_eventloop_lag_min_seconds

The minimum recorded event loop delay.

NooBaa_WebServer_nodejs_eventloop_lag_max_seconds

The maximum recorded event loop delay.

NooBaa_WebServer_nodejs_eventloop_lag_mean_seconds

The mean of the recorded event loop delays.

NooBaa_WebServer_nodejs_eventloop_lag_stddev_seconds

The standard deviation of the recorded event loop delays.

NooBaa_WebServer_nodejs_eventloop_lag_p50_seconds

The 50th percentile of the recorded event loop delays.

NooBaa_WebServer_nodejs_eventloop_lag_p90_seconds

The 90th percentile of the recorded event loop delays.

NooBaa_WebServer_nodejs_eventloop_lag_p99_seconds

The 99th percentile of the recorded event loop delays.

NooBaa_WebServer_nodejs_active_resources

Number of active resources that are currently keeping the event loop alive, grouped by async resource type.

NooBaa_WebServer_nodejs_active_resources_total

Total number of active resources.

NooBaa_WebServer_nodejs_active_handles

Number of active libuv handles grouped by handle type. Every handle type is C++ class name.

NooBaa_WebServer_nodejs_active_handles_total

Total number of active handles.

NooBaa_WebServer_nodejs_active_requests

Number of active libuv requests grouped by request type. Every request type is C++ class name.

NooBaa_WebServer_nodejs_active_requests_total

Total number of active requests.

NooBaa_WebServer_nodejs_heap_size_total_bytes

Process heap size from Node.js in bytes.

NooBaa_WebServer_nodejs_heap_size_used_bytes

Process heap size used from Node.js in bytes.

NooBaa_WebServer_nodejs_external_memory_bytes

Node.js external memory size in bytes.

NooBaa_WebServer_nodejs_heap_space_size_total_bytes

Process heap space size total from Node.js in bytes.

NooBaa_WebServer_nodejs_heap_space_size_used_bytes

Process heap space size used from Node.js in bytes.

NooBaa_WebServer_nodejs_heap_space_size_available_bytes

Process heap space size available from Node.js in bytes.

NooBaa_WebServer_nodejs_version_info

Node.js version info.

NooBaa_WebServer_nodejs_gc_duration_seconds

Garbage collection duration by kind, one of major, minor, incremental or weakcb.

NooBaa_BGWorkers_process_cpu_user_seconds_total

Total user CPU time spent in seconds.

NooBaa_BGWorkers_process_cpu_system_seconds_total

Total system CPU time spent in seconds.

NooBaa_BGWorkers_process_cpu_seconds_total

Total user and system CPU time spent in seconds.

NooBaa_BGWorkers_process_start_time_seconds

Start time of the process since unix epoch in seconds.

NooBaa_BGWorkers_process_resident_memory_bytes

Resident memory size in bytes.

NooBaa_BGWorkers_process_virtual_memory_bytes

Virtual memory size in bytes.

NooBaa_BGWorkers_process_heap_bytes

Process heap size in bytes.

NooBaa_BGWorkers_process_open_fds

Number of open file descriptors.

NooBaa_BGWorkers_process_max_fds

Maximum number of open file descriptors.

NooBaa_BGWorkers_nodejs_eventloop_lag_seconds

Lag of event loop in seconds.

NooBaa_BGWorkers_nodejs_eventloop_lag_min_seconds

The minimum recorded event loop delay.

NooBaa_BGWorkers_nodejs_eventloop_lag_max_seconds

The maximum recorded event loop delay.

NooBaa_BGWorkers_nodejs_eventloop_lag_mean_seconds

The mean of the recorded event loop delays.

NooBaa_BGWorkers_nodejs_eventloop_lag_stddev_seconds

The standard deviation of the recorded event loop delays.

NooBaa_BGWorkers_nodejs_eventloop_lag_p50_seconds

The 50th percentile of the recorded event loop delays.

NooBaa_BGWorkers_nodejs_eventloop_lag_p90_seconds

The 90th percentile of the recorded event loop delays.

NooBaa_BGWorkers_nodejs_eventloop_lag_p99_seconds

The 99th percentile of the recorded event loop delays.

NooBaa_BGWorkers_nodejs_active_resources

Number of active resources that are currently keeping the event loop alive, grouped by async resource type.

NooBaa_BGWorkers_nodejs_active_resources_total

Total number of active resources.

NooBaa_BGWorkers_nodejs_active_handles

Number of active libuv handles grouped by handle type. Every handle type is C++ class name.

NooBaa_BGWorkers_nodejs_active_handles_total

Total number of active handles.

NooBaa_BGWorkers_nodejs_active_requests

Number of active libuv requests grouped by request type. Every request type is C++ class name.

NooBaa_BGWorkers_nodejs_active_requests_total

Total number of active requests.

NooBaa_BGWorkers_nodejs_heap_size_total_bytes

Process heap size from Node.js in bytes.

NooBaa_BGWorkers_nodejs_heap_size_used_bytes

Process heap size used from Node.js in bytes.

NooBaa_BGWorkers_nodejs_external_memory_bytes

Node.js external memory size in bytes.

NooBaa_BGWorkers_nodejs_heap_space_size_total_bytes

Process heap space size total from Node.js in bytes.

NooBaa_BGWorkers_nodejs_heap_space_size_used_bytes

Process heap space size used from Node.js in bytes.

NooBaa_BGWorkers_nodejs_heap_space_size_available_bytes

Process heap space size available from Node.js in bytes.

NooBaa_BGWorkers_nodejs_version_info

Node.js version info.

NooBaa_BGWorkers_nodejs_gc_duration_seconds

Garbage collection duration by kind, one of major, minor, incremental or weakcb.

NooBaa_HostedAgents_process_cpu_user_seconds_total

Total user CPU time spent in seconds.

NooBaa_HostedAgents_process_cpu_system_seconds_total

Total system CPU time spent in seconds.

NooBaa_HostedAgents_process_cpu_seconds_total

Total user and system CPU time spent in seconds.

NooBaa_HostedAgents_process_start_time_seconds

Start time of the process since unix epoch in seconds.

NooBaa_HostedAgents_process_resident_memory_bytes

Resident memory size in bytes.

NooBaa_HostedAgents_process_virtual_memory_bytes

Virtual memory size in bytes.

NooBaa_HostedAgents_process_heap_bytes

Process heap size in bytes.

NooBaa_HostedAgents_process_open_fds

Number of open file descriptors.

NooBaa_HostedAgents_process_max_fds

Maximum number of open file descriptors.

NooBaa_HostedAgents_nodejs_eventloop_lag_seconds

Lag of event loop in seconds.

NooBaa_HostedAgents_nodejs_eventloop_lag_min_seconds

The minimum recorded event loop delay.

NooBaa_HostedAgents_nodejs_eventloop_lag_max_seconds

The maximum recorded event loop delay.

NooBaa_HostedAgents_nodejs_eventloop_lag_mean_seconds

The mean of the recorded event loop delays.

NooBaa_HostedAgents_nodejs_eventloop_lag_stddev_seconds

The standard deviation of the recorded event loop delays.

NooBaa_HostedAgents_nodejs_eventloop_lag_p50_seconds

The 50th percentile of the recorded event loop delays.

NooBaa_HostedAgents_nodejs_eventloop_lag_p90_seconds

The 90th percentile of the recorded event loop delays.

NooBaa_HostedAgents_nodejs_eventloop_lag_p99_seconds

The 99th percentile of the recorded event loop delays.

NooBaa_HostedAgents_nodejs_active_resources

Number of active resources that are currently keeping the event loop alive, grouped by async resource type.

NooBaa_HostedAgents_nodejs_active_resources_total

Total number of active resources.

NooBaa_HostedAgents_nodejs_active_handles

Number of active libuv handles grouped by handle type. Every handle type is C++ class name.

NooBaa_HostedAgents_nodejs_active_handles_total

Total number of active handles.

NooBaa_HostedAgents_nodejs_active_requests

Number of active libuv requests grouped by request type. Every request type is C++ class name.

NooBaa_HostedAgents_nodejs_active_requests_total

Total number of active requests.

NooBaa_HostedAgents_nodejs_heap_size_total_bytes

Process heap size from Node.js in bytes.

NooBaa_HostedAgents_nodejs_heap_size_used_bytes

Process heap size used from Node.js in bytes.

NooBaa_HostedAgents_nodejs_external_memory_bytes

Node.js external memory size in bytes.

NooBaa_HostedAgents_nodejs_heap_space_size_total_bytes

Process heap space size total from Node.js in bytes.

NooBaa_HostedAgents_nodejs_heap_space_size_used_bytes

Process heap space size used from Node.js in bytes.

NooBaa_HostedAgents_nodejs_heap_space_size_available_bytes

Process heap space size available from Node.js in bytes.

NooBaa_HostedAgents_nodejs_version_info

Node.js version info.

NooBaa_HostedAgents_nodejs_gc_duration_seconds

Garbage collection duration by kind, one of major, minor, incremental or weakcb.

NooBaa_cloud_types

Cloud Resource Types in the System

NooBaa_projects_capacity_usage

Projects Capacity Usage

NooBaa_accounts_usage_read_count

Accounts Usage Read Count

NooBaa_accounts_usage_write_count

Accounts Usage Write Count

NooBaa_accounts_usage_logical

Accounts Usage Logical

NooBaa_bucket_class_capacity_usage

Bucket Class Capacity Usage

NooBaa_unhealthy_cloud_types

Unhealthy Cloud Resource Types in the System

NooBaa_object_histo

Object Sizes Histogram Across the System

NooBaa_providers_bandwidth_write_size

Providers bandwidth write size

NooBaa_providers_bandwidth_read_size

Providers bandwidth read size

NooBaa_providers_ops_read_num

Providers number of read operations

NooBaa_providers_ops_write_num

Providers number of write operations

NooBaa_providers_physical_size

Providers Physical Stats

NooBaa_providers_logical_size

Providers Logical Stats

NooBaa_system_capacity

System capacity

NooBaa_system_info

System info

NooBaa_num_buckets

Object Buckets

NooBaa_num_namespace_buckets

Namespace Buckets

NooBaa_total_usage

Total Usage

NooBaa_accounts_num

Accounts Number

NooBaa_num_objects

Objects

NooBaa_num_unhealthy_buckets

Unhealthy Buckets

NooBaa_num_unhealthy_namespace_buckets

Unhealthy Namespace Buckets

NooBaa_num_unhealthy_pools

Unhealthy Resource Pools

NooBaa_num_unhealthy_namespace_resources

Unhealthy Namespace Resources

NooBaa_num_pools

Resource Pools

NooBaa_num_namespace_resources

Namespace Resources

NooBaa_num_unhealthy_bucket_claims

Unhealthy Bucket Claims

NooBaa_num_buckets_claims

Object Bucket Claims

NooBaa_num_objects_buckets_claims

Objects On Object Bucket Claims

NooBaa_reduction_ratio

Object Efficiency Ratio

NooBaa_object_savings_logical_size

Object Savings Logical

NooBaa_object_savings_physical_size

Object Savings Physical

NooBaa_rebuild_progress

Rebuild Progress

NooBaa_rebuild_time

Rebuild Time

NooBaa_bucket_status

Bucket Health

NooBaa_namespace_bucket_status

Namespace Bucket Health

NooBaa_bucket_tagging

Bucket Tagging

NooBaa_namespace_bucket_tagging

Namespace Bucket Tagging

NooBaa_bucket_capacity

Bucket Capacity Precent

NooBaa_bucket_size_quota

Bucket Size Quota Precent

NooBaa_bucket_quantity_quota

Bucket Quantity Quota Precent

NooBaa_bucket_object_count

Number of objects currently stored in the bucket.

NooBaa_bucket_max_objects_quota

Maximum number of objects allowed as bucket quota.

NooBaa_bucket_max_bytes_quota

Maximum storage capacity (in bytes) permitted bucket quota.

NooBaa_resource_status

Resource Health

NooBaa_namespace_resource_status

Namespace Resource Health

NooBaa_system_links

System Links

NooBaa_health_status

Health status

NooBaa_odf_health_status

Health status

NooBaa_replication_status

Replication status

NooBaa_replication_last_cycle_writes_size

Number of bytes replicated by replication_id in last replication cycle

NooBaa_replication_last_cycle_writes_num

Number of objects replicated by replication_id in last replication cycle

NooBaa_replication_last_cycle_error_writes_size

Number of error bytes replication_id in last replication cycle

NooBaa_replication_last_cycle_error_writes_num

Number of error objects replication_id in last replication cycle

NooBaa_bucket_used_bytes

Object Bucket Used Bytes